

Khmer semantic similarity: A study on word embedding for computing semantic similarity on Khmer text

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DEDICATION

We dedicate this paper to our dear friends, mentors, advisor, and family who have been our unwavering support, guiding us in our educational journey regardless of the circumstances, through every challenge and season.

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EXECUTIVE SUMMARY

Problem	Findings	Conclusions	Recommendations
1. How does the dimension of the word representation affect the semantic similarity classification?	The researcher conduct multiple experiments on two word embedding models, which are TF-IDF and Fasttext. The experiment includes setting different dimensional reduction on each model, different weight settings, two similarity measurement (Cosine Similarity and Euclidean Distance), and sets of decision threshold which is used to classify between similar and dissimilar texts. The data used to train the word embedding is scraped from Khmer news websites, underwent back translation process to archive a standard dataset requirement, and preprocessing	Dimension of the word representation affect the computation and the accuracy of the model. In the case of TF-IDF, reducing the dimension improves the accuracy from 76.1% to 87.7% with the dimension of 145.	Regarding the similarity classification, future researchers can look for other factors that may affect the similarity measurement, and factors that can improve the performance of the embedding model.

	steps to clean the data as well.		
2. How accurate is the static word representation compare to word embedding model for the Khmer language?	When comparing the performance of the static model TF-IDF to FastText, there are three things to keep in mind which are the vector dimension, the decision threshold, and the similarity measurement. TF-IDF resulted in 87.7% accuracy having the decision threshold of 0.5, reducing the vector dimension to 145, and utilizing Cosine Similarity. Whereas, FastText showed the accuracy score of 94.5%, having the decision threshold of 0.1, dimension 100, and also using Cosine Similarity measurement.	FastText word embedding model give a slightly higher accuracy, however it represents the words with only 100 dimensions.	As for future research, more fine-tuning and experiment can be done on these two methods. However, future researchers should look into BERT or GPT for word embedding as well.
3. How can word representation be utilized for text comparison?	Word representation can be utilized for text comparison by giving vector values to terms in documents. As	Word representation can be utilized for text comparison by assigning vector value to the terms in the documents	As per the demo web application, the future development can include larger file upload that can be extracted and used

	those terms or words are represented in a vector space, measurements such as Cosine Similarity and Euclidean Distance can be utilized to compute the distance between those vectors in the vector space. The distance between those vectors can be classified into similar or dissimilar. In this study, a model using FastText for word embedding and Cosine Similarity for similarity measurement will be demonstrated as a web application.	and calculate the distance between them in order to compare.	for computing for the similar documents
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ABSTRACT

With the current exponential growth of digital data that is on the internet and databases, the responsibility of making sure that the information is well-kept and easily accessible is also getting heavier. Although there are various technologies that have been developed with the implementation of machine learning, there is still limited research on the linguistics side of the Khmer language. The purpose of this study is to investigate and experiment on two word embeddings, TF-IDF and FastText for computing Khmer text semantic similarity. This research also conducted multiple experiments on the effect of vector dimension on similarity classification and two similarity measurements with Cosine Similarity and Euclidean Distance. The evaluation metrics used are accuracy, precision, recall, and f1-score. The findings of this research showed the performance of FastText word embedding model with the accuracy rate 94.5%.

Keywords: Semantic Similarity, Natural Language Processing, TF-IDF, FastText, Word Embedding

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LIST OF ABBREVIATIONS

NLP	Natural Language Processing
RAD	Rapid Application Development
CNN	Convolutional Neural Networks
STD	Suffix Tree Document
VSD	Vector Space Model
TF-IDF	Term Frequency-Inverse Document Frequency
POC	Proof of concept
ANN	Artificial Neural Network
STS	Semantic Similarity Dataset

MATHEMATICAL NOTATIONS

Notation	Definition
$\text{Cosine}(\theta)$	Cosine Similarity: cosine of the angle between two vectors
$d(P, Q)$	Euclidean Distance: straight-line distance between two points in Euclidean space
$\ v\ $	Vector Norm: measures the length or size of a vector
$(A * B)$	Matrix multiplication: multiplication operation between two matrices, resulting in a new matrix
$(A \odot B)$	Element-wise Multiplication: multiplication of corresponding elements in two matrices of the same size
$(A \cdot B)$	Vector dot product
(A^T)	Transpose of a matrix
$\frac{d}{dx}(f)$	Derivative

Σ	Summation
$\log()$	Logarithm function
e	The base of natural logarithm
$\in$	Set membership: indicates that an element is a member of a set
$s(w, n)$	Similarity score between the target word w and a negative example n

Chapter 1: The Problem and Its Scope

1.1 Introduction

In the digital transformation era, Artificial Intelligence has become a driving force that reshapes industries and research domains worldwide. This revolution of AI enables numerous autonomous performance of various tasks, and one of them is semantic similarity. The significance discovery of this semantic similarity model has enabled the computational analysis of the likeness between text.

Semantic similarity measure has great importance in natural language processing (NLP) and many computer applications related field such as text evaluation, information retrieval, text classification, text summarization, and question answering [1].

1.2 Rationale of the Study

The task of identifying the similarity between text can be done manually for a small set of articles, but when it comes to dealing with larger set of data, it becomes a challenge that requires heavy manual labor and time consumption. Therefore, semantic similarity tools are crucial for efficiently and accurately analyzing text.

Currently, there are several researches done on computing the semantic similarity between text and its evolution throughout the years. However, those studies are designed for more commonly spoken languages such as English, Chinese, German, French, and Japanese. Meanwhile, the study on semantic information and tools for the Khmer text are still understudied, due to the fact that the Khmer language is a low-resources language [2].

Therefore, a research and development on the semantic similarity of Khmer language is crucial for the development of technology for Khmer language and for the

study of natural language processing. In this case, the research will be focusing on the different word embedding for returning the word vectors that will be used to calculate the similarity.

1.3 Statement of the Problem

This research aims to study on word embeddings TF-IDF and FastText for computing the semantic similarity of Khmer language. Furthermore, this study seeks to answer the following questions:

1. How does the dimension of the word representation affect the semantic similarity classification?
2. How accurate is the static word representation compare to word embedding model for the Khmer language?
3. How can word representation be utilized for paragraph comparison?

1.4 Significance of the Study

The significance of Khmer document similarity analysis lies in its ability to unlock a new way to study linguistic patterns and improve digital knowledge retrieval for documents written in Khmer text. The study of semantic similarity provides many extensive use cases, especially in clustering [3] [4], and many different algorithms and models have been proposed in the English language and have achieved outstanding results. Nevertheless, when it comes to a language as complex as the Khmer language, no standard or benchmark Khmer language dataset can be used to create a model that fits precisely for this [5].

Nonetheless, this study will be beneficial to a variety of fields that deal with a large amount of information or documents that are written in the Khmer language, including but not limited to:

- Development of Khmer language processing tools: stemming from understanding the semantic information of Khmer text, various tools can be developed for processing Khmer text. Those tools include sentiment text analysis and translation and are more tailored to Khmer script.
- Enhancing search engine capabilities for Khmer text: understanding the semantic information of Khmer text will leave space for improvement of search engines on the internet. By refining the ability to interpreting the text correctly, it can ensure accurate and relevant search outcomes based on the information provided by the user while using the search engine.

1.5 Scope and Limitation

The study will focus on researching and comparing between two word embedding models, which are TF-IDF and Fasttext, for embedding words and returning vectors that can be used to computing the semantic similarity between Khmer news article. As the model will be designed to deal with Khmer text, it would not be able to process text written in other languages. Likewise, the model's primary objective is to provide the vector value of the Khmer words that will be used to compute the similarity by calculating cosine similarity or Euclidean distance.

There are a few limitations to this research. Firstly, the lack of standardized Khmer text corpus [5]. Secondly, the proposed research does not cover other problems of Khmer text, such as incorrect grammar and incorrect spelling. Therefore, the input is expected to be written in Khmer and with correct spelling.

1.6 Definition of Terms

Natural Language Processing: is a field of artificial intelligence that focuses on the interaction between computers and human language. It enables machines to understand, interpret, and generate human language, which develop into applications such as semantic similarity analysis, sentiment analysis, and more

Semantic Similarity: refers to the degree of likeness or resemblance in meaning between two or more pieces of text or data. It quantifies how closely related or similar the underlying concepts or content of the text are.

Source Document: refers to a text or document from which information or content is extracted or compared. In the context of semantic similarity, the source document is typically the reference text or document that serves as the basis for measuring the similarity between the given documents.

Target Document: is the document being compared or evaluated in relation to the source document. In this context, the target document is the text or document that is compared to the source document to determine how similar or related they are.

Vector Space Model: A conceptual model in which text documents are represented as vectors in a multidimensional space. Each term, sentence, and doc will be represented in a shared space that will be used to compute the semantic similarity between those vectors.

Graph Neural Networks: A machine learning approach that focuses on processing graph-structure data. In this study, the text or document will be represented in as a node in a graph. GNN will be utilized to create a new representation that factor in the node that are connected by being under the same topic.

Suffix Tree Document: A data structure employed in text processing and pattern matching, representing various possible suffixes of a string.

Term Similarity Weighting Tree: A structure used in computational linguistics, designed to weight terms based on their semantic similarity within a specific context.

Bidirectional Long Short Term Memory: An enhancement of the LSTM model, augmenting text data comprehension through processing in both forward and reverse directions.

Long Short Term Memory: A variant of recurrent neural networks, employed in deep learning. It exhibits significant efficacy in learning order dependencies in sequence prediction.

1.7 Theoretical Framework

In order to explore this issue of computing the similarity between Khmer text documents deeply, several theories and studies that have been published and related to this study should be discussed. The first section of this theoretical framework concerns the development of the prototype that will be used for demonstration, and the main process that the research will go through while researching. Secondly, the data for training and evaluating the model; the third section provides an overview of the process of training an NLP model; and the last section shows the framework for testing the model.

1.7.1 Software Development Life Cycle (SDLC)

The Software Development Life Cycle (SDLC) represents a structured approach designed to deliver superior software while optimizing cost and development duration. Since this is a study on machine learning, the prototype application will follow an SDLC called AI Project Life Cycle.

AI Project Lifecycle

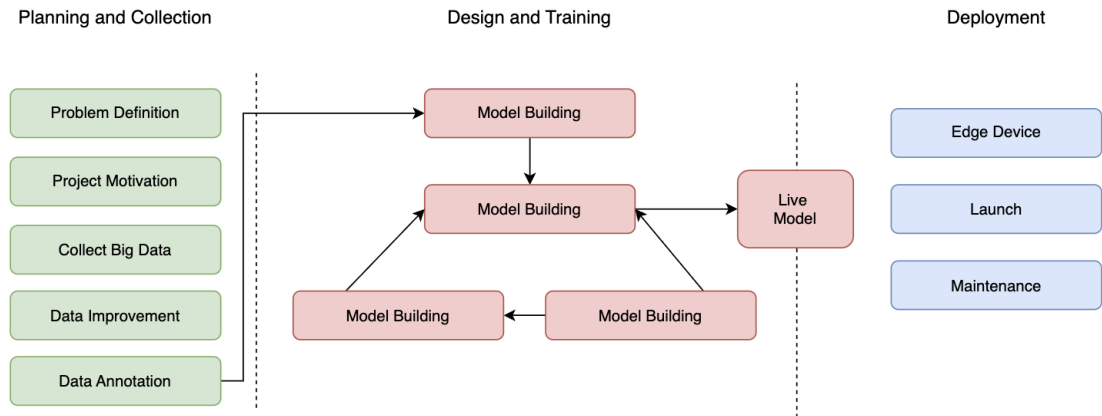


Figure 1: AI Project Lifecycle

The AI project lifecycle [6] involves several stages, each critical for the successful development and deployment of AI models. In this project, this SDLC is utilized to develop and do experiment on the Khmer semantic similarity models from the start to finish. This cycle comprises three main steps, which are:

Planning and Collection

- Problem definition: Identify and define the problem related to Khmer text similarity that the project aims to solve.
- Project Motivation: Outline the reasons and benefits behind researching on the Khmer semantic similarity.
- Collect Big Data: Gather dataset to Khmer texts such as news articles to build a comprehensive dataset for training and evaluating the model
- Data Improvement: Clean and preprocess the collected data to insure its quality and consistency.
- Data Annotation: Label the data with the correct class (similar or not similar) for training and evaluation

Design and Training

- **Model Building:** Develop different types of model to compute the similarity between the Khmer texts. The types of model are according to the frameworks.
- **Evaluation:** Assess the performance of the models using predefined metrics such as the accuracy, precision, recall, and F1-score, and comparing the performance between the models.
- **Refinement:** Based on the evaluation result, refine and improve the models to increase the performance.
- **Training:** Train and refined models on the improvement and the dataset.

Deployment

- **Live Model:** Deploy the trained models into a production environment where they can be accessed and utilized by end-users.
- **Edge Devices:** If applicable, deploy the model on a mobile device for on-the-go text comparison.
- **Launch:** Release the application to the user to test and use.
- **Maintenance:** continuously monitored and maintain the deployed models to ensure their performance and address any issues.

By following this AI software development lifecycle, the development of the model and the prototype application is well-structured.

1.7.2 NLP Training Framework

Gearing towards the NLP method, the theoretical framework that will be used to demonstrate the details on how the researcher is going to build, train and test the semantic similarity model. This section includes, the data generation process,

similarity measurement using TF-IDF and similarity measurement using FastText.

Those processes are represented in the following:

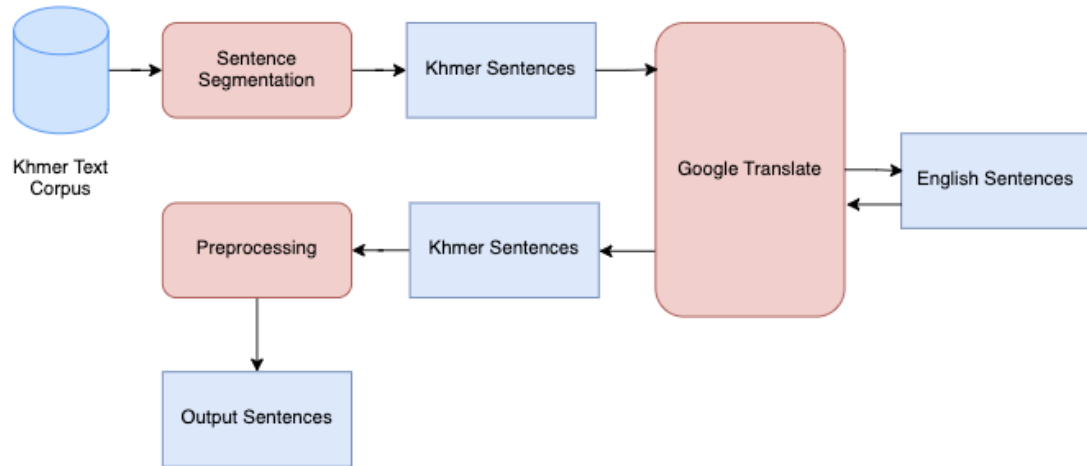


Figure 2: Data Generation Framework

Figure 2 demonstrates the process of generating a set of datasets from the existing Khmer news article dataset. To train the model, a dataset with sentences pair that has almost the same meaning is needed [5]. The process starts by segmenting the articles to a sentence level by utilizing the Python library, khmer-nltk. Next, the output sentences will undergo a back translation process, using Google Translation, to English and back to Khmer, resulting in another Khmer sentence that has the very similar meaning to the original sentence but written in slightly different way. The output sentence, which is the pair sentence, will go through a preprocessing step that ensure the data quality.

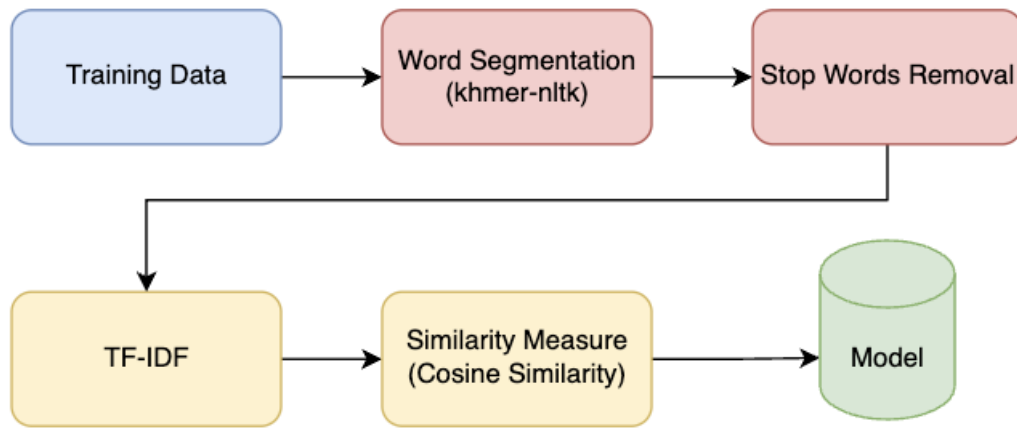


Figure 3: Training Framework using TF-IDF

Figure 3 illustrates the training process of the NLP model used to build the semantic similarity measurement using the traditional method to evaluate the relevance of words, TF-IDF. The diagram shows the flow of the process, beginning with a collection of data that will be processed. The initial processing utilizes khmer-nltk word segmentation, a crucial part of processing and dealing with Khmer text when it comes to language processing in machine learning. The following processing step is removing the stop words. According to [6], a stop word refers to a word in a language commonly used and often eliminated when it comes to text processing, as it carries very little helpful information. Upon completion of these processing steps, the words in the dataset will be ranked based on the statistical calculation of TF-IDF. The following step, cosine similarity, will be used to calculate the similarity between those vectorized text, resulting in a model that will serve as a baseline for the semantic similarity.

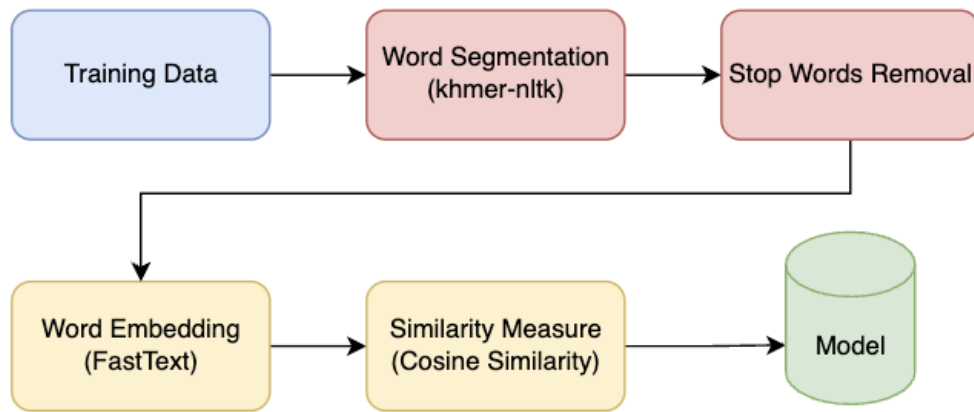


Figure 4: Similarity Training Framework using FastText

Figure 4 demonstrates a similar process of computing the similarity between Khmer text. Firstly, the training data undergo the same preprocessing steps which is the word segmentation, and stop word removal. Secondly, the output word will be passed to a word embedding algorithm that will provide a vector representation of the data. The next step, cosine similarity, will be used to calculate the similarity between those embedding words.

1.7.3 NLP Testing Framework

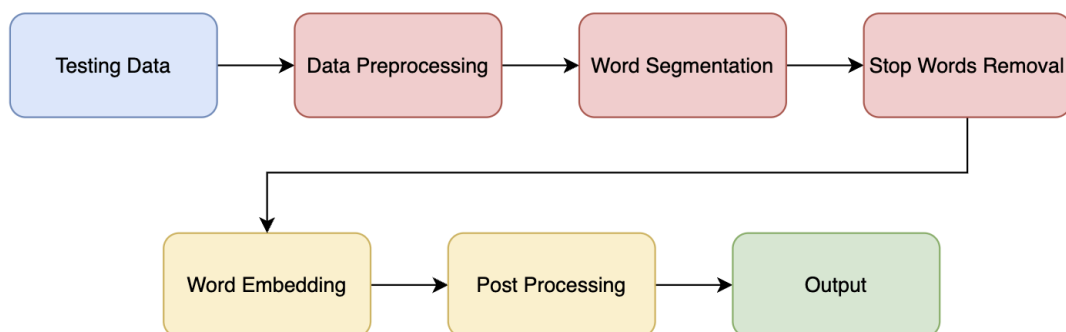


Figure 5: Semantic Similarity Testing Framework

In Figure 4, the process of testing the NLP model is shown in a step-by-step procedure, starting from the preprocessing of the test data, preparing it for the word segmentation process, and removing stop words from the data. Once the data processing step is completed, the word embedding process will start by taking the resulting data from the processing step and embedding it into vector space. Upon the completion of this essential step, the by-product, which is the vectorized data, is passed to the post-processing stage, where the model will calculate the semantic similarity and output the degree of how similar the input texts are.

1.7.4 Software Testing Framework

Unified Theory of Acceptance and Use of Technology

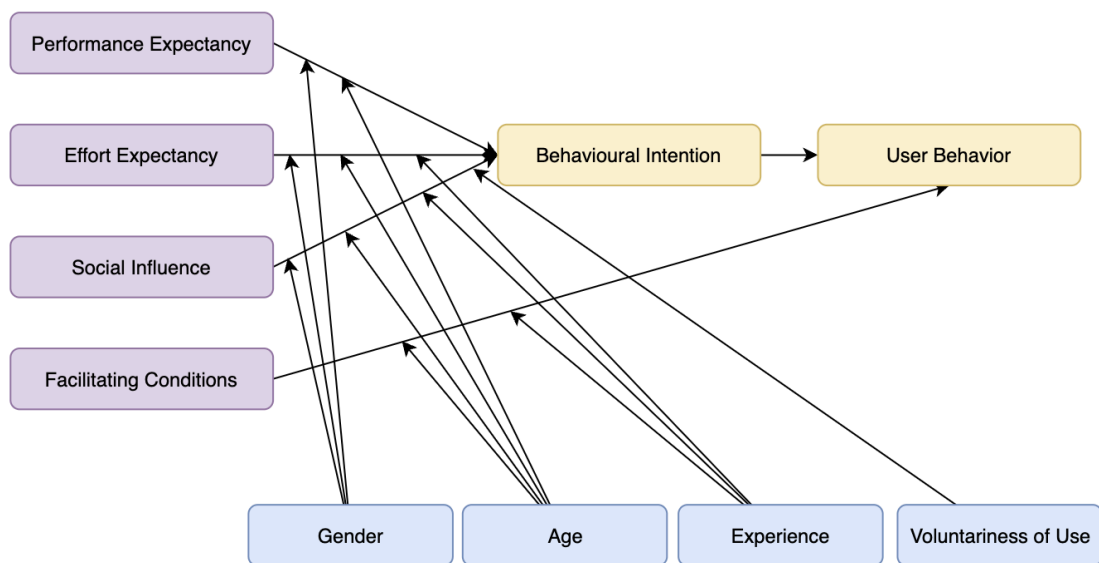


Figure 6: Unified Theory of Acceptance and Use of Technology

The Unified Theory of Acceptance and Use of Technology (UTAUT) [7] is a model that aims to explain user intention to use an information system and the usage behavior. Developed by Venkatesh et al. in 2003, UTAUT integrates elements from eight prominent models of technology acceptance to provide a comprehensive

framework. It identifies four key constructs that influence user acceptance and usage behavior. Those four key elements are:

- **Performance Expectancy:** is the degree to which an individual believes that using the system will help them in their work. The higher the performance means that the user have more intention to use the system.
- **Effort Expectancy:** This refers to the degree of ease associated with the use of the system. It derived from perceived ease of use and complexity from Technology Acceptance Model (TAM), Theory of Planned Behaviour (TPB), and the Model of PC Utilization (MPCU).
- **Social Influence:** is defined as the degree that which an individual is influenced by other people to use the system. This includes the user that uses the system due to mandatory requirements and not personal preferences. In this case, social influence measure the degree that the user uses the Khmer semantic similarity as a requirement and a part of their work.
- **Facilitating Condition:** refers to the degree in which an individual believes that an organization would support the use of the system.

1.8 Conceptual Framework

Furthermore, to better explain the progression of this study, this section provides a visual representation of how research or testing will be done, as well as details on the approach to be carried out.

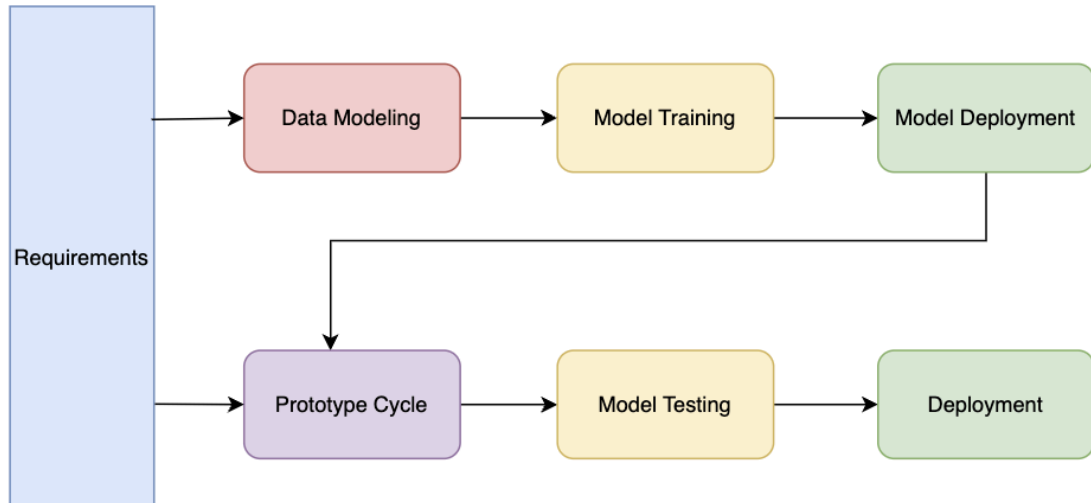


Figure 8: Semantic Similarity System

Figure 8 demonstrates the entire structure of the entire system for both the model and the prototype or the proof of concept design by the researcher. The framework briefly demonstrates the process by showing two builds using the same SDLC model, RAD, from the initial stage of requirement gathering until the deployment of the POC.

The initial stage of the research begins with gathering the requirements for the model training and the prototype. In this stage, the researcher starts by looking at various approaches that have been studied and analyzes the required data, as well as any input and output that were used in those processes. The first stage of this research will be mainly focusing on the NLP model development. Following this, the data needed for the experiment will be gathered and preprocessed to ensure its quality. After the data is processed, the next stage will be training the model, where the expected outcome is an NLP model that can process and compute the semantic information of Khmer text and output the degree of similarity between them.

For the second iteration of the research, the focus will be shifting from developing an NLP model to the development of the prototype or the proof of concept

(POC) that will be used to demonstrate the model. In this build, more attention will be paid to the development of the POC, which will also lead to the completion of the testing phase. Lastly, the system will be prepared for deployment.

Chapter 2: Literature review

2.1 Review of Related Literature

Currently, there are existing models on Hugging Face that are being used to capture semantic information that is useful for various usage such as semantic search, text comparison, clustering, and other sentence similarity related tasks. Those models include:

all-MiniLM-L6-v2 [8] this model aims to train sentence embedding models on very large sentence level datasets using a self-supervised contrastive learning objective. This model used the pretrained nreimers/MiniLM-L6-H384-uncased [9] model and fine-tuned in on a 1B sentence pairs dataset, including a contrastive learning objective: given a sentence from the pair, the model should predict which out of a set of randomly sampled other sentences, was actually paired with it in the dataset.

all-mpnet-base-v2 [10] used a pre-trained microsoft/mpnet-base [11] model and fine-tuned in on a 1B sentence pairs dataset, using a contrastive learning objective: given a sentence from the pair, the model should predict which out of a set of randomly sampled other sentences, was actually paired with it in our dataset.

Although there are a few existing technologies that are developed for comparing the similarity between texts, it is important to note that those models are not designed for the complexity of Khmer language. Therefore, it is crucial that there should be models designed and trained for Khmer texts.

2.2 Review of Related Studies

In the presence of Khmer, semantic similarity is a topic that is also being studied, but there are still some limitations and areas that need to be explored and

researched. However, there is some research concerning this topic, as well as studies in the neighboring languages, that can be studied and experimented on.

In [12], the authors introduced two methods for calculating the similarity between two news articles that are in two different languages, Hindi and English. They divided it into two main processes. Firstly, the news articles in both English and Hindi that were referring to the same event were extracted from Google's news feed, and the Hindi news article was translated into English using Google Translate and then compared with English news articles. Secondly, they used Cosine similarity, Jaccard similarity, and Euclidean distance measure to calculate the news articles similarity score. The underlying concept of this paper is to identify feature article vectors and calculate the difference between those feature vectors; having a low distance between those features implies a high level of similarity value, while a significant distance implies a low level of similarity value. All three methods experimenting on both TF-IDF vectors and bags of words showed that cosine similarity using TF-IDF had a greater accuracy, recall, and F-measure score than the other methods mentioned. The other two methods were also stated to be improved with the Doc2Vec model, which takes corpus as input and generates document vectors as output [13].

Another study on a hybrid model [14] aims to tackle the problem of understanding the meaning of terms and using a traditional similarity calculation method called term frequency-inverse document frequency (TF-IDF). The model mentioned in this paper is independent of a large corpus to support it. However, this study adopted a complete dictionary as their data source for dealing with the meaning of the words. Moreover, they also utilize the term similarity weighting tree structure (TSWT) and semantic information from HowNet. The result of this study mentioned

how the model decreases computing time, shows a better clustering method, and improves the accuracy over the traditional methods.

In regard to the FastText word embedding model, a study on sentiment analysis of COVID-19 [15] highlighted the effectiveness of using FastText as a word embedding model. Its proficiency in embedding words based on subword information demonstrates how it tailored specifically to the dataset's characteristics and outperform the traditional method TF-IDF and another embedding model, GLoVe.

Moreover, in this study [16], the researchers also discussed the performance of FastText in comparison to other embedding techniques such as Word2Vec and GloVe. They emphasized on how these model performs differently in the context of a morphologically complex language, which FastText were able to slightly outperform the other two techniques.

With the recent development in neural networks, various studies have been conducted, and new approaches in language processing have been proposed with neural network techniques, including Convolutional Neural Networks (CNN), Long Short Term Memory (LSTM), Bidirectional Long Short Term Memory (Bi-LSTM), Artificial Neural Network (ANN) [1].

In one study on Khmer semantic similarity, S. Houn [5], proposed a deep learning model for sentence vectorization and comparison by utilizing different processes of data preprocessing on Khmer text, including text segmentation, stop-words removal, translating the text to and from English, resulting in an output sentence that is used as a dataset that is used to train as well as evaluate their model. They proposed a deep-learning approach that is both used for sentence vectorization, Bi-LSTM and text comparison that utilize Artificial Neural Networks (ANN) to

classify the sentences into two classes, 1 (similar) and 0 (dissimilar), based on a predefined threshold, achieving an accuracy of 84.86%.

2.3 Review of Related Technologies

2.3.1 khmer-nltk

Khmer-nltk [17] is a Python library that provide tools for natural language preprocessing. The library offers tools for the Khmer text such as sentence segmentation, word segmentation, and part-of-speech tagging. With the limited available tools, khmer-nltk is tailored and to handle the processing of the Khmer text very well.

2.3.2 Jinja

Jinja [18] is a fast, expressive, and extensible template engine for Python designed to generate dynamic content for web applications. Jinja seamlessly embedded Python code within the templates, facilitating the integration of ML models. This template engine will ease the process of development of the prototype for semantic similarity.

2.3.3 Python

Python [19] is a widely used programming language that is known for its simplicity, readability, and vast variety of frameworks and libraries. When it comes to testing and developing a semantic similarity model, this programming language offers several advantages such as its tools and seamless integration.

2.3.6 FastAPI

FastAPI [20] is a modern and high-performance web framework for building API with Python, with automatic interactive document. This Khmer semantic similarity system will use FastAPI to create a web service, allowing the user to upload text and receive similarity results via a web interface.

2.3.7 Scikit-learn (sklearn)

Scikit-learn [21] is an open-source machine learning library for Python providing different tools and algorithms for data analysis, classification, regression, or clustering. In this case, in Khmer semantic similarity, it uses the TF-IDF vectorizer model, TfidfVectorizer for building the baseline model.

2.3.8 SciPy

SciPy [22] is an open-source Python library used for scientific and technical computing. It builds on NumPy and other functions that operate on NumPy arrays. In this project, SciPy provides the similarity measurement functions, Cosine Similarity and Euclidean Distance.

2.3.9 Gensim

Gensim [23] is an open-source Python library designed for topic modeling and document similarity analysis using various statistical machines learning techniques, known for its scalability and efficiency in handling large text corpora.

Chapter 3: Research Methodology

3.1 Research Design

This study is an applied research with a primary focus on experimental methodologies, experimenting with different approaches intending to explore and test various techniques for assessing semantic similarity between Khmer text.

Moreover, this research a quantitative research with a primary focus on computing the similarities between the Khmer text. According to this paper, quantitative research is an approach of investigating certain phenomena using statistical and numerical data. In this study, the process of calculating the similarity between text and the evaluation of the model involve numerical analysis like the accuracy, precision, recall, and F1-Score.

3.2 Methodology

Moreover, after reviewing the related literature and technologies, there are many solutions that solve the problem of computing the semantic information of texts. Each of the methods has its own advantages and disadvantages, but hybrid methods and deep neural networks approach seem to perform better than the traditional methods of comparing words. The main process of this semantic similarity model is data preprocessing and word embedding.

3.3.1 Data Collection

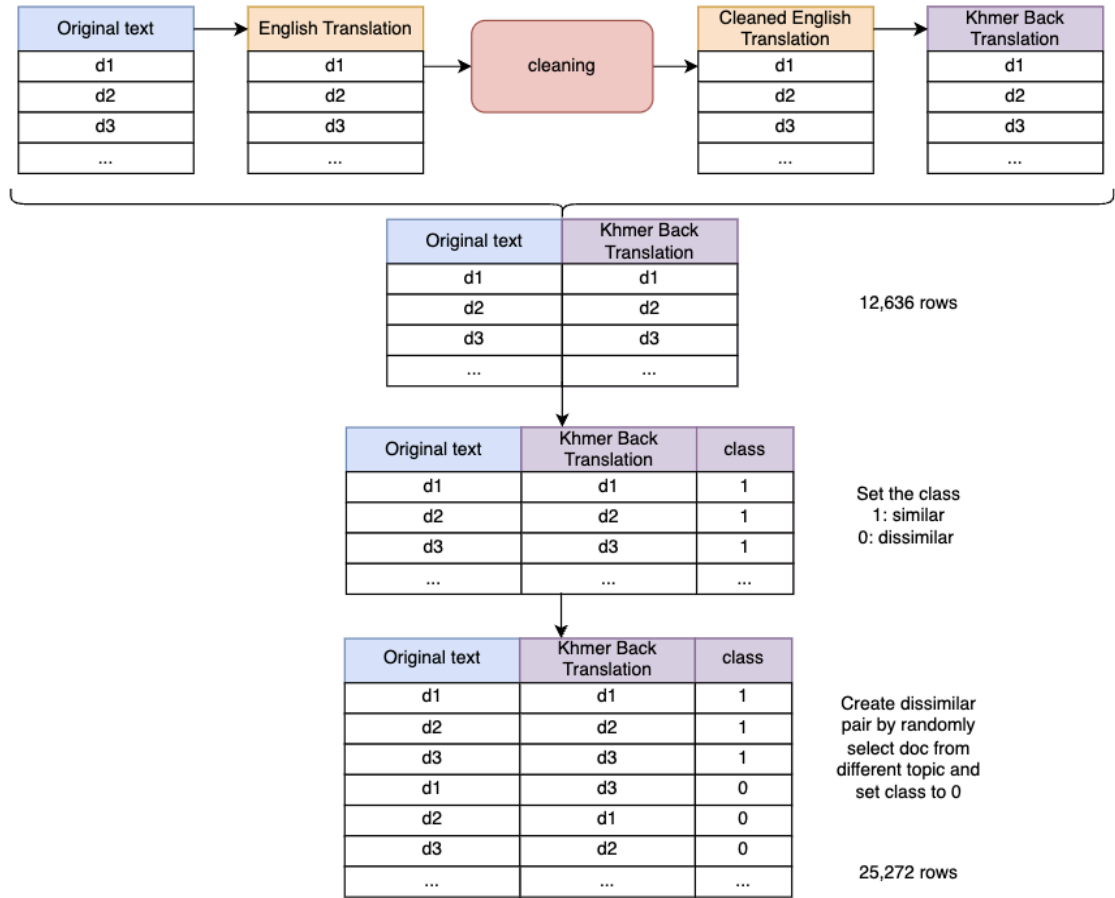


Figure 8: Process of Dataset Creation

The dataset will be used in this research is a corpus of Khmer news articles that was scraped from Khmer news websites such as BTV [24], Khmerload [25], Sabaynews [26], B8-sport [27], Popular [28]. This dataset consists of 20 thousand news articles with its respective title, topic, filename, and URL. This collection provides a diverse set of texts necessary for training and evaluating the semantic similarity model. Each entry of the dataset includes the following attributes as shown in the table:

- title: the headline or the title of the news article
- topic: the category or subject of the news article

- filename: the name of the file to retrieve the news article
- url: the web address from which the article was scraped.

title	topic	filename	url
មកដឹងពីគុណសម្បត្តិ និងគុណវិបត្តិ រថយន្តបំពាក់ស៊ីម៉ង់ត៍៖	auto talk	SB-auto-talk-1 683165855886 022000.txt	https://news.sabay.com.kh/article/1040610
មនុស្ស ៩ នាក់ ស្លាប់ក្រោយលេចធ្លាយឧស្ម័ន នៅរដ្ឋ Punjabប្រទេសឥណ្ឌា	ព័ត៌មាន អន្តរជាតិ	btv-international-168310607 9382711600.txt	https://btv.com.kh/article/22783
សៀ សុផារ៉ា បង្ហាញមុខលើកនេះប្លែកខ្លាំង សម្រស់ដូចកាលទើបលឿងបូងៗ	តារា	popular-celebrity-168405464 9408719328.txt	https://www.popular.com.kh/entertainment/695395.html/
ពានរង្វាន់សម្តេចហ៊ុន សែនថ្នាក់ខេត្តឆ្នាំនេះមាន២ ០ក្រុមចូលរួម	កីឡា	b8-sport-1683 913295337429 267.txt	https://www.b8-sport.net/archives/sports-news/6861
ប្រុងប្រយ័ត្នផង! បេក្ខភាពត្រូវប្រឡងបាក់ខុប ជៀសវាងករណីទាំងនេះ ប្រយ័ត្នត្រូវលុបបេក្ខភាពប្រឡង	យល់ដឹង	KL-knowledge-1682348217 571421128.txt	https://www.khmerload.com/article/197198

Table 1: Raw Dataset

The column ‘title’, ‘topic’, and ‘url’ are dropped as it is not needed, and the column filename is changed to ‘source’. This ‘source’ column indicates the filename of the original text, which can also be used to access the whole text file.

source
SB-auto-talk-1683165855886022000.txt
btv-international-1683106079382711600.txt
popular-celebrity-1684054649408719328.txt
b8-sport-1683913295337429267.txt
KL-knowledge-1682348217571421128.txt

Table 2: Example of Filename in the Dataset

The next step, of creating the dataset, is to create a pair article for each news article by performing a process called back translation. Back translation or reverse translation is a process of translating a previously translated text back to its source language. Firstly, in the initial translation step, each news article is translated to English text by utilizing Googletrans, which is a public library from PyPi. This step helps to create a reference version of the articles in an understood language, which the researcher can check the result and make sure it is in an acceptable quality. As shown in the table 2, the Khmer news article is translated to English, and there are a few errors such as duplicated word, numbers, characters, and no space between the full stop “.” and the new sentence. These errors will be eliminated and processed in the next step before translated back to the original language, which is the Khmer language.

Original Text	English Translated Text
វេបសាយវិភាគ FiveThirtyEight បានចេញផ្សាយការព្យាករណ៍ Premier	Analysis website FiveThirtyEight has released updated Premier League

League ដែលបានធ្វើបច្ចុប្បន្នភាពបន្ទាប់ពីការប្រកួតកាលពីចុងសប្តាហ៍មុន។ Man United មិនបានចូលរួមក្នុងលីគកាលពីចុងសប្តាហ៍មុនទេ ដូច្នេះតំណែងរបស់គេមិនមានការផ្លាស់ប្តូរខ្លាំងទេ។ ពួកគេនៅតែជាក្រុមមួយក្នុងចំណោមក្រុមបីដែលមានឱកាសជាក់ស្តែងក្នុងការឈ្នះការប្រកួត ប៉ុន្តែពួកគេគឺជាក្រុមខាងក្រៅទៅវិញក្នុងការប្រកួតប្រជែងលីគ នេះជាមួយនឹងលទ្ធភាពបីភាគរយឈរនៅលើកំពូលតារាងនៅទីបញ្ចប់រដូវកាល។ Arsenal គឺជាក្រុមដែលមានភាគរយខ្ពស់ជាងគេខណៈដែលអ្វីៗស្ថិតនៅដដែល។ ឱកាសឈ្នះពានលីគរបស់ពួកគេត្រូវបានគេរាយការណ៍ថាមាន៥៣ភាគរយធៀបនឹងឱកាស៤៤ភាគរយដែល Man City ឈ្នះវាម្តងទៀត។ នៅថ្ងៃអាទិត្យបន្ទាប់ Man United នឹងប្រកួតនៅកីឡដ្ឋាន Anfield។ ប្រសិនបើពួកគេយកឈ្នះ Liverpool នោះពួកគេអាចបង្កើតឱកាសខ្លួនឯងបានយ៉ាងល្អជាកម្លាំងជំនួយឱ្យទុកចិត្តនៅកំពូលតារាងនៅចុងរដូវកាល។ ប្រភព៖ FiveThirtyEight	predictions after last weekend's game. Man United did not join the league last weekend, so his position has not changed much. They are still one of three teams with a real chance of winning the match, but they are the only team outside of the league with a 3% chance of standing at the top of the table at the end of the season. Arsenal are the team with the highest percentage while everything stays the same. Their chances of winning the league are estimated at 53%, compared to Man City's 44% chance of winning it again. Next Sunday, Man United will play at Anfield. If they beat Liverpool, they can create a good chance of themselves as a reliable force at the top of the table at the end of the season. Source: FiveThirtyEight
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Table 3: Example of English Translation of Khmer Text

In this next step, the uncleaned translated text that contains the elements such as duplicated word, numbers, characters, and no space between the full stop “.” and the new sentence. This has found to affect the result of the back translation process, which is the pair text of each article. Therefore, it is crucial to make sure that the English text is cleaned of those issues before translating it back to the original text. The cleaning process consists of removing any duplicated word, number, character, double spaces, and that there is one space in front of each new sentence. The table 4, shows the result and comparison between uncleaned and cleaned text.

Uncleaned English Translated Text	Cleaned English Translated Text
Analysis website FiveThirtyEight has released updated Premier League predictions after last weekend's game.Man United did not join the league last weekend, so his position has not changed much.They are still one of three teams with a real chance of winning the match, but they are the only team outside of the league with a a 3% chance of standing at the top of the table at the end of the season.Arsenal are the team with the highest percentage while everything stays the the same.Their chances of winning the league are estimated at 53%, compared to Man City's 44% chance of winning it it again.Next Sunday, Man United will play at Anfield.If they beat Liverpool, they can create a a good chance of	Analysis website FiveThirtyEight has released updated Premier League predictions after last weekend's game. Man United did not join the league last weekend, so his position has not changed much. They are still one of three teams with a real chance of winning the match, but they are the only team outside of the league with a 3% chance of standing at the top of the table at the end of the season. Arsenal are the team with the highest percentage while everything stays the same. Their chances of winning the league are estimated at 53%, compared to Man City's 44% chance of winning it again. Next Sunday, Man United will play at Anfield. If they beat Liverpool, they can create a good chance of themselves as a

themselves as a a reliable force at the top of the table at the end of the season.Source: FiveThirtyEight	reliable force at the top of the table at the end of the season. Source: FiveThirtyEight
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Table 4: Example of Cleaning the Translated English Text

After the completion of the initial translation and cleaning, the English text undergoes a back translation process, which is translating the English text back to Khmer text. This back translated process produced paired articles that maintained the original meaning of the original text, but had slight variations in the wording and phrasing. These paired articles are essential for training models to recognize semantic similarity despite the differences in expression. The result of back translated text is shown in the table 5, and the comparison between the source text (the original text) and the target text (the paired text), is shown in table 6. Furthermore, both the source and target text are segmented word by word, utilizing khmer-nltk.

Cleaned English Translated Text	Back Translated Text to Khmer
Analysis website FiveThirtyEight has released updated Premier League predictions after last weekend's game. Man United did not join the league last weekend, so his position has not changed much. They are still one of three teams with a real chance of winning the match, but they are the only team outside of the league with a 3% chance of standing at the top of the table at the end of the season. Arsenal are the team with the highest percentage while everything stays the same. Their chances of winning the league are estimated at 53%, compared to Man City's 44% chance of winning it again. Next Sunday, Man United will play at Anfield. If they beat Liverpool, they can create a good chance of themselves as a reliable force at the top of the table at the end of the season. Source: FiveThirtyEight	គេហទំព័រវិភាគ FiveThirtyEight បានចេញផ្សាយការព្យាករណ៍ Premier League ដែលបានធ្វើបច្ចុប្បន្នភាពបន្ទាប់ពីការប្រកួតកាលពីចុងសប្តាហ៍មុន។ Man United មិនបានចូលរួមលើកកាលពីចុងសប្តាហ៍មុនទេ ដូច្នេះតំណែងរបស់គេមិនមានការផ្លាស់ប្តូរប្រើនទេ។ ពួកគេនៅតែជាក្រុមមួយក្នុងចំណោមក្រុមទាំងបីដែលមានឱកាសពិតប្រាកដក្នុងការឈ្នះការប្រកួត ប៉ុន្តែពួកគេគឺជាក្រុមតែមួយគត់ដែលនៅក្រៅលីគដែលមានឱកាស 3% ក្នុងការឈរនៅលើកំពូលតារាងនៅចុងរដូវកាលនេះ។ Arsenal គឺជាក្រុមដែលមានភាគរយខ្ពស់ជាងគេខណៈដែលអ្វីៗនៅដដែល។ ឱកាសឈ្នះពានលីករបស់ពួកគេត្រូវបានគេប៉ាន់ប្រមាណថាមាន ៥៣% បើធៀបនឹងឱកាស ៤៤% របស់ Man City ក្នុងការឈ្នះវាម្តងទៀត។ នៅថ្ងៃអាទិត្យក្រោយ Man United នឹងលេងនៅកីឡដ្ឋាន Anfield។ ប្រសិនបើពួកគេយកឈ្នះ Liverpool ពួកគេអាចបង្កើតឱកាសដ៏ល្អមួយសម្រាប់ខ្លួនពួកគេក្នុងនាមជាកម្លាំងដែលអាចទុកចិត្តបាននៅលើកំពូលតារាងនៅចុងបញ្ចប់នៃរដូវកាលនេះ។ ប្រភព៖

	FiveThirtyEight
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Table 5: Example of Back Translation

Source (Original Text)	Target (Pair Text)
វេបសាយវិភាគ FiveThirtyEight បានចេញផ្សាយការព្យាករណ៍ Premier League ដែលបានធ្វើបច្ចុប្បន្នភាពបន្ទាប់ពីការបុ រក្ខតកាលពីចុងសប្តាហ៍មុន។ Man United មិនបានចូលរួមក្នុងលីគកាលពីចុងសប្តា ហ៍មុនទេ ដូច្នេះតំណែងរបស់គេមិនមានការផ្លា ស់ប្តូរខ្លាំងទេ។ ពួកគេនៅតែជាក្រុមមួយក្នុងចំណោមក្រុ មបីដែលមានឱកាសជាក់ស្តែងក្នុងការ ឈ្នះការប្រកួត ប៉ុន្តែពួកគេគឺជាក្រុមខាងក្រៅទៅវិញ ក្នុងការប្រកួតប្រជែងលីគ នេះជាមួយនឹងលទ្ធភាពបីភាគរយឈរ នៅលើកំពូលតារាងនៅទីបញ្ចប់រដូវកា ល។ Arsenal គឺជាក្រុមដែលមានភាគរយខ្ពស់ជាងគេ ខណៈដែលអ្វីៗស្ថិតនៅដដែល។ ឱកាសឈ្នះពានលីគរបស់ពួកគេត្រូវ បានគេវាយតម្លៃថាមាន៥៣ភាគរយ ធៀបនឹងឱកាស៤៤ភាគរយដែល Man City ឈ្នះវាម្តងទៀត។ នៅថ្ងៃអាទិត្យបន្ទាប់ Man United នឹងប្រកួតនៅកីឡដ្ឋាន Anfield។ ប្រសិនបើពួកគេយកឈ្នះ Liverpool	គេហទំព័រវិភាគ FiveThirtyEight បានចេញផ្សាយការព្យាករណ៍ Premier League ដែលបានធ្វើបច្ចុប្បន្នភាពបន្ទាប់ពីការបុ រក្ខតកាលពីចុងសប្តាហ៍មុន។ Man United មិនបានចូលរួមលីគកាលពីចុងសប្តាហ៍ មុនទេ ដូច្នេះតំណែងរបស់គេមិនមានការផ្លា ស់ប្តូរប្រើនទេ។ ពួកគេនៅតែជាក្រុមមួយក្នុងចំណោមកុ រុមទាំងបីដែលមានឱកាសពិតប្រាកដក្នុ ងការឈ្នះការប្រកួត ប៉ុន្តែពួកគេគឺជាក្រុមតែមួយគត់ដែល នៅក្រៅលីគដែលមានឱកាស 3% ក្នុងការឈរនៅលើកំពូលតារាងនៅចុង រដូវកាលនេះ។ Arsenal គឺជាក្រុមដែលមានភាគរយខ្ពស់ជាងគេ ខណៈដែលអ្វីៗនៅដដែល។ ឱកាសឈ្នះពានលីគរបស់ពួកគេត្រូវ បានគេប៉ាន់ប្រមាណថាមាន៥៣% បើធៀបនឹងឱកាស៤៤%របស់ Man City ក្នុងការឈ្នះវាម្តងទៀត។ នៅថ្ងៃអាទិត្យក្រោយ Man United នឹងលេងនៅកីឡដ្ឋាន Anfield។ ប្រសិនបើពួកគេយកឈ្នះ Liverpool ពួកគេអាចបង្កើតឱកាសដ៏ល្អមួយសម្រាប់

នោះពួកគេអាចបង្កើតឱកាសខ្លួនឯង បានយ៉ាងល្អជាកម្លាំងដ៏គួរឱ្យទុកចិត្ត នៅកំពូលតារាងនៅចុងរដូវកាល។ ប្រភព៖ FiveThirtyEight	រាប់ខ្លួនពួកគេក្នុងនាមជាកម្លាំងដែលអា ចទុកចិត្តបាននៅលើកំពូលតារាងនៅច ុងបញ្ចប់នៃរដូវកាលនេះ។ ប្រភព៖ FiveThirtyEight
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Table 6: Example of an Original Text and Its Pair

Source (Original Text)	Target (Pair Text)
វេបសាយវិភាគ FiveThirtyEight បាន ចេញផ្សាយ ការព្យាករណ៍ Premier League ដែល បាន ធ្វើ បច្ចុប្បន្នភាព បន្ទាប់ពី ការប្រកួត កាលពី ចុង សប្តាហ៍ មុន ។ Man United មិន បាន ចូលរួម ក្នុង លីគ កាលពី ចុង សប្តាហ៍ មុន ទេ ដូច្នេះ តំណែង របស់ គេ មិន មាន ការផ្លាស់ប្តូរ ខ្លាំង ទេ ។ ពួកគេ នៅតែ ជា ក្រុម មួយ ក្នុង ចំណោម ក្រុម បី ដែល មាន ឱកាស ជាក់ស្តែង ក្នុង ការឈ្នះ ការប្រកួត ប៉ុន្តែ ពួកគេ គឺជា ក្រុម ខាងក្រៅ ទៅវិញ ក្នុង ការប្រកួតប្រជែង លីគ នេះ ជាមួយនឹង លទ្ធភាព បី ភាគរយ ឈរ នៅលើ កំពូល តារាង នៅ ទីបញ្ចប់ រដូវកាល ។ Arsenal គឺជា ក្រុម ដែល មាន ភាគរយ ខ្ពស់ ជាង គេ ខណៈ ដែល អ្វី ៗ ស្ថិត នៅ ដដែល ។ ឱកាស ឈ្នះ ពាន លី គរបស់ ពួកគេ ត្រូវបាន គេ វាយតម្លៃ ថា មាន ៥៣ ភាគរយ ធៀប នឹង ឱកាស ៤៤ ភាគរយ ដែល Man City ឈ្នះ វា ម្តង ទៀត ។ នៅ ថ្ងៃអាទិត្យ បន្ទាប់ Man United នឹង ប្រកួត នៅ កីឡដ្ឋាន Anfield ។ ប្រសិនបើ ពួកគេ យកឈ្នះ Liverpool នោះ ពួកគេ អាច បង្កើត ឱកាស ខ្លួន ឯង បាន យ៉ាង ល្អ ជា កម្លាំង ដ៏ គួរឱ្យ ទុកចិត្ត នៅ កំពូល តារាង នៅ ចុង រដូវកាល ។ ប្រភព៖ FiveThirtyEight	គេហទំព័រវិភាគ FiveThirtyEight បាន ចេញ ផ្សាយ ការ ព្យាករណ៍ Premier League ដែល បាន ធ្វើ បច្ចុប្បន្ន ភាព បន្ទាប់ ពី ការ ប្រកួត កាល ពី ចុង សប្តាហ៍ មុន ។ Man United មិន បាន ចូលរួម លីគ កាលពី ចុង សប្តាហ៍ មុន ទេ ដូច្នេះ តំណែង របស់ គេ មិន មាន ការផ្លាស់ប្តូរ ច្រើន ទេ ។ ពួកគេ នៅតែ ជា ក្រុម មួយ ក្នុង ចំណោម ក្រុម ទាំង បី ដែល មាន ឱកាស ពិត ប្រាកដ ក្នុង ការឈ្នះ ការប្រកួត ប៉ុន្តែ ពួកគេ គឺជា ក្រុម តែ មួយ គត់ ដែល នៅក្រៅ លីគ ដែល មាន ឱកាស 3 % ក្នុង ការឈរ នៅ លើ កំពូល តារាង នៅ ចុង រដូវកាល នេះ ។ Arsenal គឺជា ក្រុម ដែល មាន ភាគរយ ខ្ពស់ ជាងគេ ខណៈ ដែលអ្វី ៗ នៅ ដដែល ។ ឱកាសឈ្នះ ពាន លីគ របស់ ពួកគេ ត្រូវបាន គេ ប៉ាន់ប្រមាណ ថា មាន ៥៣ % បើ ធៀប នឹង ឱកាស ៤៤ % របស់ Man City ក្នុង ការឈ្នះ វា ម្តង ទៀត ។ នៅ ថ្ងៃអាទិត្យ ក្រោយ Man United នឹង លេង នៅ កីឡដ្ឋាន Anfield។ ប្រសិនបើ ពួកគេ យកឈ្នះ Liverpool ពួកគេ អាច បង្កើត ឱកាស ដ៏ ល្អ មួយ សម្រាប់ ខ្លួន ពួកគេ ក្នុង នាម ជា កម្លាំង ដែល អាច ទុកចិត្ត បាន នៅ លើ កំពូល តារាង នៅ ចុងបញ្ចប់ នៃ រដូវកាល នេះ ។ ប្រភព ៖ FiveThirtyEight

Table 7: Example of the Segmented Original Text and Its Pair

Before the completion of creating the dataset, each pair of text is given a class to indicate the similarity between the two texts. The class consists of two categories, which are “1”: “similar” and “0”: “not similar”.

The class for all the original text and its pair are set to “1”, indicating the similarity. The news article with pair text that are not similar are created by randomly selecting the news article that is not under the same category (sport, international, auto-talk, social, celebrity, knowledge), and assigned the class to “0”, indicating not similar.

source	target	class
KL-sport-1682293164890179217.seg.txt	KL-sport-1682293164890179217.pair.seg.txt	1
btv-international-1683106341740226700.seg.txt	btv-international-1683106341740226700.pair.seg.txt	1
b8-sport-1683917176132275591.seg.txt	btv-international-1683106320542752600.seg.txt	0
SB-social-1683207054335381402.seg.txt	btv-sport-1683169178959063000.seg.txt	0

Table 8: The Resulted Dataset

In the next step of preparing the data for training and evaluating the model, a further preprocessing step is conducted to increase the quality of the available dataset. The objective of this next preprocessing step is to remove special characters, symbols, and emoticons. The preprocessing step is crucial in training any NLP models, this step

includes word segmentation where sentences will be separated into each individual word and stored as a list. The next step is removing stop words, where predefined stop words that have little contextual meaning will be removed.

Character Removal

This part of the pipeline further clean the dataset of characters that are not needed and ensures that the input text is clean, standardized, and free of noise. Breaking down the cleaning process in details: specific characters, punctuations, digits, and other symbols are removed as it are non-informative and do not contribute to the meaning of the text but can interfere with the text processing. Moreover, extra spaces are condensed into a single space, non-breaking spaces are replaced with regular space, and English characters are converted to lowercases. Characters that were removed are listed down in the table below:

Type	Element
Characters	[/ #] () “ « » , . ? - ... + =
Khmer Punctuations	្ក ្ខ ្គ ្ឃ ្ង ្ច ្ឆ ្ជ ្ឈ ្ញ
Lunar Date Symbols	☐ ☐
Digits	០ ១ ២ ៣ ៤ ៥ ៦ ៧ ៨ ៩ ០ ១ ២ ៣ ៤ ៥ ៦ ៧ ៨ ៩

Table 9: List of Removed Characters

Word Segmentation

The first key point in preprocessing text is segmentation. In the task of computing the semantic similarity, each word needs to be considered for its contextual meaning. However, the Structure of the Khmer language does not make it simple when it comes to segmenting words, like for example, English language, where each

word is separated by space with a clear delimiter. An example of separating the sentences into words are:

- ខ្ញុំញ៉ាំបាយ (I eat rice) $\Rightarrow$ ខ្ញុំ (I) + ញ៉ាំ (eat) + បាយ (rice)

Stop word removal

In analyzing the semantic similarity between text, each word is taken into consideration based on its meaning. Stop words are words that appear very frequently and are insignificant to identifying text content. When it comes to computing the semantic similarity of text, stop words will lead to significant error, which is why they are being treated as noises and being removed. The deletion of the stop word will be achieved by a predefined list of words. The table below shows the list of stop word in consideration:

សព្វនាម	"ខ្ញុំ" "អ្នក" "នាង" "ពួក" "យើង" "ពួកគេ" "លោក"
ឃ្លាភ្ជាប់	"និង" "ដែល" "ជា" "តែ" "ដើម្បី" "ប៉ុន្តែ" "ដោយសារ" "ពេលដែល" "ហើយ" "ដូចជា" "ដូច្នេះ" "ពេលណាមួយ" "ទៅវិញ" "តែម្តង" "ជាមួយ" "ដូចគ្នានឹង" "រួចហើយទេ"

	“ဗု” “မေတ္တာ”
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Table 10: List of Stop Words

3.3.3 Word Embedding

TF-IDF

TF-IDF, stands for Term Frequency-Inverse Document Frequency, is a statistical method used to evaluate the importance of a word in a corpus. In computing the similarity between text, TF-IDF method is a traditional statistics-based text similarity measure algorithm, which constructed model by text word frequency vector [13]. TF-IDF is used to assess how important a word is in a document in a collection or corpus. Thus, helping in checking with documents are similar based on the word usage. A word in the document is considered as a term, and the weight is assigned to each term based on how frequently it appears in the corpus. The TF-IDF score for a word is calculated as follows:

- **Term Frequency (TF):** measures how frequently a term occurs in a document

$$TF(w, d) = \frac{\text{Total number of words in document } d}{\text{Number of times word } w \text{ appears in document } d}$$

- **Inverse Document Frequency (IDF):** measures how important a term is within the entire corpus

$$IDF(w) = \log\left(\frac{\text{Total number of documents}}{\text{Number of documents containing word } w}\right)$$

- **TF-IDF:** the product of TF and IDF, giving a weight to each term in the document

$$TF - IDF(w, d) = TF(w, d) \times IDF(w)$$

Where:

- w is the term
- d is the document

In this case, the number of total document is equal to the number of all the original text, which is 12,636 document. The size of this matrix, which is the number of unique word, is equal to 145. An example of how words are represented using TF-IDF:

Term	Document 1	Document 2		Document n
ការឈ្នះ	0.0023	0		0.005
ការប្រកួត	0.0041	0.003		0.017
ចំនោម	0.001	0		0
ឱកាស	0.102	0		0
....				

Table 11: Example of Word Representation with TF-IDF

FastText

FastText is an extension of word2vec model that was developed by Facebook AI Research (FAIR) for understanding of word representations and sentence classification [29]. FastText is an unsupervised embedding model that only requires to be segmented corpus, and can be trained in two ways, such as skipgram and cbow (continuous-bag-of-words) [30]. In the case of semantic similarity, it has been proven to perform better than the traditional TF-IDF.

The foundation of FastText is the skip-gram model introduced by [31] and this model learns the word representation by predicting the given target word by observing the given a context word. The objective of the model is to maximize the

log-likelihood of observing both context word and target word. This simply means that it will sum all the log probabilities of all context words w_c given the target word w_t across the entire corpus.

$$\sum_{t=1}^T \sum_{c \in C_t} \log p(w_c | w_t)$$

Where:

- T is the total number of words in the training corpus
- C_t is the set of context words for the target words w_t
- w_t is the target word
- w_c is the context word within the context window around w_t

The probability of a context word w_c , given a target word w_t , is computed using the softmax function:

$$p(w_c | w_t) = \frac{e^{s(w_t, w_c)}}{\sum_{j=1}^W e^{s(w_t, j)}}$$

Where:

- $p(w_c | w_t)$ is the probability of context word w_c given a target word w_t
- $s(w_t, w_c)$ is the scoring function based on the word vectors
- W is the size of the vocabulary

The scoring function between the context word w_c and the target word w_t :

$$s(w_t, w_c) = u_{w_t}^\top v_{w_c}$$

Where:

- s refers to the scoring function
- u_{w_t} is the vector representation for the target word
- v_{w_c} is the vector representation for context word

Due to the computational complexity of the softmax function, which requires the summation of all words in the vocabulary, FastText uses negative sampling to simplify the optimization. This function transform multi-class classification into multiple binary classification problem. This function maximizes the probability of the actual context word and minimizes the probability of randomly sampled negative words.

$$\log(1 + e^{-s(w_t, w_c)}) + \sum_{n \in N_{t,c}} \log(1 + e^{s(w_t, n)})$$

Where:

- $N_{t,c}$ is a set of negative examples sampled from the vocabulary
- n is each word in the set of negative sample

By denoting the logistic loss function $l: x \rightarrow \log(1 + e^{-x})$, the main objective function can be rewritten as:

$$\sum_{t=1}^T \left[\sum_{c \in C_t} l(s(w_t, w_c)) + \sum_{n \in N_{t,c}} l(-s(w_t, n)) \right]$$

In the paper [fasttext paper], the FastText model incorporates the subword information by representing each word as a bag of character n-grams. This means that each word is broken into n number of characters, with a special boundary symbols < and > added to the beginning and the end of the word to distinguish prefixes and suffixes.

Taking a word “flower” and n-gram value $n = 3$ as an example, the word (w) “flower” will be represented by the character n-gram:

<fl, flo, low, owe, wer, er>

And the special sequence: <flower>

The sequence <low> or <owe>, corresponding to the word *low* and *owe* is different from the tri-gram *low* and *owe*.

The difference between the general skip-gram model and FastText model, is the scoring function. Therefore, supposed that a given dictionary of n-gram of size G , $G_w \subset \{1, \dots, G\}$ representing n-gram of w .

$$s(w, c) = \sum_{g \in G_w} z_g^T v_c$$

Where:

- s refers to the scoring function
- g is the n-gram representation of each target word
- z_g is the vector representation of each g (n-gram)
- v_c is the vector representation of the context word
- G_w is the dictionary of n-gram

An example of how words are represented using FastText:

Words	Dimension 1	Dimension 2		Dimension 100
ការឈ្នះ	0.23	0.11		-0.33
ការប្រកួត	0.25	-0.30		0.28
ចំនោម	0.40	0.09		-0.28
ឱកាស	-0.35	0.20		0.05
....				

Table 12: Example of Word Representation with Fasttext

3.3.4 Similarity Measure

Similarity measurement is one of the main component of this study as they take in the embedded words and analyse the similarity of them. There are various of methods ranging from the mathematical equation to measure the distance of the vectors embedded in the vector space, to using neural network to perform text classification.

In this study, the researcher will look into three methods that are often used to compute for the similarity between vectors, and analyse the one that has the highest performance for the Khmer language. Those three methods include Cosine similarity, Euclidean Distance, and Graph Neural Network.

Cosine Similarity

Cosine similarity is a mathematical method used to compute the similarity between two non-zero vectors in a multidimensional space [32]. This equation compute the cosine of an angle between two text resulting in a value range of -1 to 1, where -1 indicates complete similarity, and 1 indicates complete similarity [33].

$$\text{Cosine Similarity} = 1 - \frac{A \cdot B}{||A|| ||B||} = 1 - \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}}$$

Where:

- A_i and B_i are components of vectors A and B, respectively

Euclidean Distance

Euclidean distance is a measuring method that used to calculate distance between two points as a straight line in Euclidean space. Euclidean space is a plane where each of the points is defined by a set of coordinates which determine the position of the point in space relative to a fixed reference point [34]. In this study, a text is represented as a node of a graph in vector space, therefore, the placement of the nodes can affect the distance that is supposed to be measured for its similarity. With the spatial properties of graph in concerned, Euclidean distance is a more suitable approach than Cosine similarity as it measures the straight line between one point to another.

Euclidean distance formula calculate the distance between two points A and B with their respective coordinates, $(a_1, a_2, a_3, ..., a_n)$ and $(b_1, b_2, b_3, ..., b_n)$, using this formula:

$$d(P, Q) = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2 + \dots + (a_n - b_n)^2}$$

3.3.5 Evaluation

In order to evaluate the outcome of this computation, the most popular evaluation metrics such as accuracy, precision, recall, F1-score will be used. However, human evaluation will also be needed when it comes to evaluation of the Khmer text.

In this case, some articles will be chosen from the test data and determine the similarity and dissimilarity by human inspection [12]. This judgement will be used as a benchmark for testing the automatic similarity calculation on the Khmer text.

Accuracy

Accuracy measures the ratio of the correct predicted value over the total number of instance evaluated [35]. In this task, accuracy reflects the percentage of times the model correctly identifies similar or dissimilar text pairs.

		Actual Value	
		Positive	Negative
Predicted Value	Positive	TP	FP
	Negative	FN	TN

Figure 9: Confusion Matrix

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

- *TP* (True Positives) is the actual similar texts correctly identified as similar
- *TN* (True Negatives) is the actual dissimilar text correctly identified as dissimilar
- *FP* (False Positives) is the actual similar text incorrectly identified as dissimilar

- *FN* (False Negatives) is the actual dissimilar text incorrectly identified as similar

Precision

Precision is a measurement of the positive pattern that are correctly predicted from the total predicted pattern in the positive class. It is the result of dividing the True Positive value by the confusion matrix's total True Positive and False Positive values. In this case, precision focuses on how many times the model correctly predict the pairs that are actually similar.

$$Precision = \frac{TP}{TP + FP}$$

Recall

Recall is used to measure the fraction of positive pattern that are correctly classified. It is the result of dividing the True Positive value of a confusion matrix by the total of the True Positive and False Negative values. In this study, recall takes the opposite approach from precision.

$$Recall = \frac{TP}{TP + FN}$$

F1-Score

F1-Score is derived by multiplying a confusion matrix's recall and accuracy by the sum of recall and precision. It shows how well a method can balance precision and recall, and it measures the overall effectiveness of the method. In this study, F1-score indicates the model's performance on identifying truly similar pairs and avoiding false positives.

$$F1 - Score = 2 * \frac{Precision * Recall}{Precision + Recall}$$

3.3 Prototype

To demonstrate the model performance with real data, the researcher will create a search engine web application to accept the input of a Khmer news article with its topic and output the related news articles.

Regarding the front-end development, the researcher will use Python and Jinja [18] to develop a web application to demonstrate the model usage and performance.

For the back-end development, the researcher will use FastAPI [20] to develop the API endpoints to accept and return request from the client side.

The functionality of the finished prototype would be as follows:

1. The user fills in the input with Khmer text and its topic
2. The user sent the request to the server
3. The server process the request and return a response with a list of documents semantically similar to the input document
4. The application display the list of semantically similar document

3.6 Statistical Treatment of Data

In this research, statistical method like descriptive statistic is important when it comes to analyzing and interpreting the data collected. The descriptive statistic methods that are crucial in understanding the data such as mean, median, mode, standard deviation, variance [36].

Mean can be used to calculate the average length of articles and sentences in the dataset. This provides insight into the general length size of the article or sentence length.

Median will be used to determine the middle value in the sorted list that has the article or sentence length, which will provide a better representation of the dataset.

Mode will be used to identify the most frequently occurring article and sentence length in the dataset that will show the most common characteristics in the dataset.

Standard Deviation can be applied to measure the variability in length of the sentence and article in the dataset. This information can be used to assess the performance on the length.

Variance can provide a clear measure of how much an individual article and sentence length differ from the average in the dataset. This is essential for designing the semantic similarity model for the dataset.

Chapter 4: Presentation, Analysis and Interpretation of Data

4.1 Description of the findings for SOP 1

How does the dimension of the word representation affect the semantic similarity classification?

Reduction in dimension in this study means reducing the size of the vector representation of text, while attempting to preserve as many key features as possible. By reducing the dimension, less important information is removed and also helps to reduce the noise in the data.

In answering this question, 10 experiments have been conducted on the word representation TF-IDF using Cosine Similarity and Euclidean Distance as similarity measurements. Moreover, each measurement has an optimal decision threshold that is used to classify the text, whether it is similar or dissimilar. For Cosine similarity, if the similarity measurement is less than 0.5, the texts are classified as similar, and dissimilar otherwise. Whereas, for Euclidean Distance, the texts are classified as similar, if the similarity measurement is less than 0.1, and dissimilar otherwise. The result shows varying performances in accuracy, precision, recall, and F1-score as the dimension decreases.

This decrease in the dimension is the result of limiting the unique vocabulary in the word representation by ignoring any word that has the number of occurrences below the set configuration. In this experiment, 'min_df' was used to reduce the dimension. The 'min_df' is used for removing terms or words that appear too infrequently in the dataset. For example:

- $\text{min_df} = 0.01$ means “ignore terms that appear in less than 1% of the documents in the dataset”
- $\text{min_df} = 0.1$ means “ignore terms that appear in less than 10% of the documents in the dataset”

TF-IDF					
Type: Dimensional Reduction					
Similarity Measurement: Cosine Similarity					
Decision Threshold = 0.5					
Experiment	Accuracy	Precision	Recall	F1-Score	Dimension
0: $\text{min_df}=0$	76.1%	100%	52.3%	68.7%	48796
1: $\text{min_df}=0.01$	83.7%	100%	67.5%	80.5%	1636
2: $\text{min_df}=0.02$	84.9%	100%	69.8%	82.2%	923
3: $\text{min_df}=0.1$	87.7%	97.3%	77.6%	86.3%	145
4: $\text{min_df}=0.2$	82.3%	82%	82.7%	82.4%	46
Similarity Measurement: Euclidean Distance					
Decision Threshold = 1.3					
Experiment	Accuracy	Precision	Recall	F1-Score	Dimension
0:	98.7%	99.8%	97.6%	98.7%	48796

min_df=0					
1: min_df=0.01	98%	97.5%	98.6%	98%	1636
2: min_df=0.02	95.7%	93.6%	98.1%	95.8%	923
3: min_df=0.1	73.3%	65.9%	96.4%	78.3%	145
4: min_df=0.2	60.3%	56%	95.5%	70.6%	46

Table 13: Dimensionality Reduction Experiments

For Cosine Similarity measurement with the decision threshold of 0.5, the performance in terms of accuracy and F1-score improves as the dimension decrease from the original dimension, 48796 to 145. This is the result of removing any term that appear in less than 10% of the documents. However, it drops when the dimension is 46.

Whereas for Euclidean Distance, the highest performance of F1-score and accuracy of 98% at 1636 dimensions. This suggests that this measurement may require higher dimension to effectively capture the similarity between text more accurately.

4.2 Description of the findings for SOP 2

How accurate is the static word representation compare to word embedding model for the Khmer language?

A comparison between a static word representation which is the TF-IDF and word embedding models, which is FastText, has been determined by the similarity measurement using the Cosine Similarity and Euclidean Distance. Each of the word

embedding models is evaluated under the set decision threshold that would optimize the performance when computing for the similarity.

As shown in the table below, the performance of the TF-IDF embedding measured by Cosine Similarity has the accuracy of 87.7%. This result is achieved by evaluating on pairs of documents that comprise both the similar and dissimilar pairs, experiment on dimensional reduction, and set an optimal decision threshold. Having the precision of 97.3%, it shows the percentage of the model correctly classified the pairs that are actually similar. Whereas, when it comes to identify all the truly similar pairs out of all that existed, the model has the recall value of 77.6%. The overall performance on identifying both the truly similar and incorrectly label dissimilar pairs is 86.3%. The same experiments and processes were being done when calculating similarity using Euclidean Distance, achieving the accuracy of 94.7%.

On the other hand, the result from FastText model when utilizing Euclidean Distance to calculate the similarity, gives a rather less accurate result when comparing to measurement using Cosine Similarity. The accuracy score from the FastText model using Cosine Similarity is 94.5% and the F1-score of 94.3%. As compare to the highest performance from the TF-IDF, FastText model uses vector of 100 dimension to represent the terms, where TF-IDF uses 145 dimensions.

However, it can be seen that various of experimentation were performed on TF-IDF model and the performance as well as the dimension size of the vector representation is still larger than FastText. Therefore, FastText achieve a higher accuracy and able to have a smaller vector representation size than TF-IDF.

Word embedding	Configuration	Similarity measurement	Accuracy Score	Precision	Recall	F1 score
TF-IDF	-Decision	Cosine	87.7%	97.3%	77.6%	86.3%

	threshold: 0.5, -min_df=0.1	Similarity				
	-Decision threshold: 1.3 -min_df=0.01	Euclidean Distance	98%	97.5%	98.6%	98%
FastText	Decision threshold: 0.1	Cosine Similarity	94.5%	97.8%	91.1%	94.3%
	Decision threshold: 0.5	Euclidean Distance	82.6%	99.8%	65.3%	78.9%

Table 14: Comparison between TF-IDF and FastText

4.3 Description of the findings for SOP 3

How can word representation be utilized for text comparison?

To answer this question, it is crucial to consider the process when it comes to comparing two texts. There are two factors to keep in mind, which are the word representation and the comparison metrics. Word representation can be utilized for text comparison by giving vector values to terms in documents. As those terms or words are represented in a vector space, measurements such as Cosine Similarity and Euclidean Distance can be utilized to compute the distance between those vectors in the vector space. The distance between those vectors can be classified into similar or dissimilar.

In this study, two word embeddings were experimented and compared, and the results are shown above. Even though, TF-IDF has shown the high performance when it comes to embedding Khmer text to use for computing the similarity, it is important to be mindful of the size of the vector as well, which is the dimension. Based on the result of the experiments, FastText performs well with the accuracy score of 94.5%

and F1-score of 94.3%, and resulting in a vector size of 100 dimension. Moreover, it is evident that Cosine Similarity was utilized to measure the similarity score between texts, which effectively determining how similar the texts are.

This FastText model was trained and evaluated on the dataset that consists of Khmer news articles, therefore, its usage will be demonstrated as a web application. Additionally, the demonstration of the usage will be shown as comparing a text to the existing news articles in the database, by also utilizing Cosine Similarity as the similarity measurement. The top 10 articles that are similar to the input text will be displayed on the interface. Therefore, word representation can be utilized for text comparison by assigning vector value to the terms in the documents and calculate the distance between them in order to compare.

Word embedding	Configuration	Similarity measurement	Accuracy Score	Precision	Recall	F1 score
FastText	Decision threshold: 0.1	Cosine Similarity	94.5%	97.8%	91.1%	94.3%

Table 15: FastText Model for Demo App

Chapter 5: Technical Aspect of the KSS

5.1 Software Requirements Specification (SRS)

5.1.1 Purpose

The purpose of this section is to list all the software requirements specifications for the Khmer Semantic Similarity. This section composes of general design constrains and system requirements.

5.1.2 General Design Constrains

Application Environment

This research is solely on semantic similarity on Khmer language. The components of this are different word embedding models and similarity measurement using different methods, which are cosine similarity and Euclidean distance. To interact with these models, a FastAPI backend server is developed and act as a web service to receiving user's requests which are Khmer text to calculate similarity to find the similar documents that are in the database. A simple web application utilizing Jinja framework is developed to interact with the backend. Therefore, the user can interact with the web application interface to get a list of documents that are semantically similar to their input text.

User Characteristics

This study focuses on researching and developing a Khmer semantic similarity model and compare between different embedding methods. The main target users are the all users who want to do researchers and developers. This research can be a stepping stone for researcher and developer to expand their knowledge on Khmer

semantic similarity, make use of different models for comparing Khmer texts, and expand the system into a desirable product fitting for their usage.

Mandated Constrains

For this research, the functionality and usage of the models are demonstrated through a simple web application. This system contains some mandated constrains, as listed down below.

- **Language:** this system is built for processing and handling Khmer text
- **Technology:** this research is built on the backend server FastAPI, therefore, if the application experiences downtime, the users will be temporary unable to use the system.

5.1.3 System Requirement

Functional Requirement

The functional requirement details on the functionalities of the Khmer semantic similarity system, and all the activities that users of this system can perform.

ID	REQUIREMENT	DESCRIPTION
FR1	Upload the document	The system should allow the user to upload the document that they want to search from.
FR2	Upload the text	The system allow the user to upload the text string that they want to search from.
FR3	Obtain the result	The system should display a list of document that has been found to be similar to the uploaded document or text.

Table 16: Functional Requirements

Non-Functional Requirement

The non-functional requirement describes the characteristics and the limitations of the application and the expectations of the users.

ID	REQUIREMENT	DESCRIPTION
NFR1	Security	The system does not store any input document and input text
NFR2	Intuitiveness	The application for demonstration offers an intuitive and uncomplicated interface, allowing users to easily navigate and use the application

Table 4: Non-functional Requirements

Hardware Requirements of Development Process

The computer specifications that the researchers used to build and train the Khmer semantic similarity models are available on Google Colab, and they consist of the following:

- Architecture: NVIDIA Ampere
- CUDA Cores: 1,0240 CUDA Cores
- GPU Memory: 56 GB
- Memory Bandwidth: 1.5 TB/s

Performance Requirements

- **Text Input Handling:** the system should display the entered text in the text box instantaneously, and the text submission should trigger the embedding and similarity calculation within 2 seconds
- **Result Display:** the system should calculate the similarity and display the list of similar documents within 2 seconds

- **User Interface Responsiveness:** the user interface should remain responsive and interactive throughout the user's interaction

User Interface

The demo for the model in this research will be demonstrated through a simple web application interface, that was developed using Jinja, which is a Python framework, and FastAPI for the backend server.

Software Interface

User can use the Khmer semantic similarity models by interact with the application that is used to demonstrating the models. The demo application user must interact is the web application interface.

Operating Environment

Client/Server System

Database: Supabase

Processing Unit: M1 Pro

Operating System: macOS Sonoma

Platform: Jinja + FastAPI

Interface Design



Figure 10: Interface Design

5.2 Software Design and Architecture (SDA)

5.2.1 High-Level Components Relation Diagram

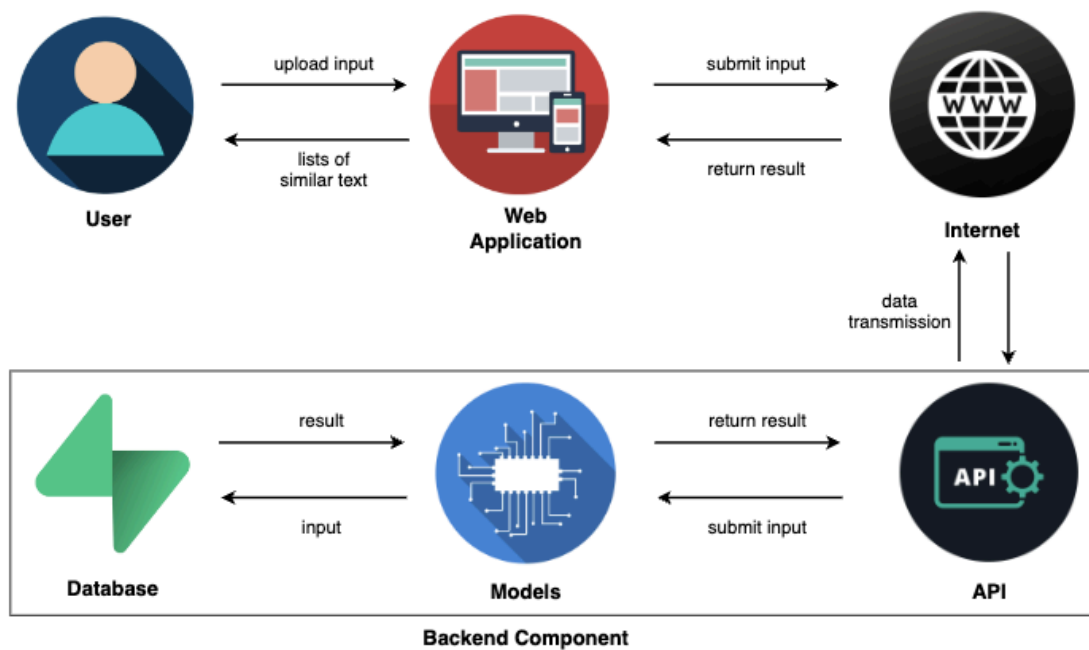


Figure 11: High-Level Components Relation Diagram

Figure NUMBER illustrates the high level component relationship between all the components in the Khmer semantic similarity system. It allows a user to input Khmer text and receive a list of the top 10 most similar text articles from a database. The web application frontend interact with users and communicates their input to the backend via the internet. The backend, consisting of an API, processing the models, a Supabase database performing word embedding and similarity measurement and return the result. The components consist in this diagram are:

- User: represent a user who inputs Khmer text and receives a list of similar text articles as output.
- Web Application: provide the interface for user to interact, and interface for displaying the results back to the users.
- Internet: facilitates data transmission between the web application and the backend server.
- API: received input from the web application, send it to the model, and send the result back to the web application.
- Models: refers to the word embedding models (TF-IDF, Fasttext), perform the word embedding and similarity calculations, interact with the database to retrieve the stored text articles.
- Database: store the text articles used for similarity comparison, provide data to the model for processing.

5.2.2 Use Case View

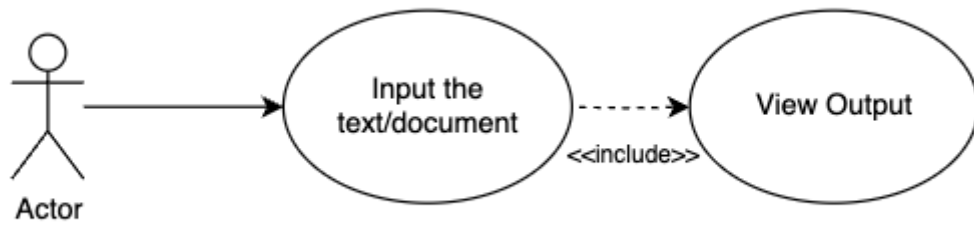


Figure 11: Use Case Diagram

Figure NUMBER illustrates the use case diagram of the system, showing the interactions between the user and the system. The user selects a word embedding models (TF-IDF, Fasttext) and similarity measurements methods (cosine similarity and euclidean distance). Then they input text or documents for analysis and view the output, which is the top most similar text articles based on the input text.

5.2.3 Component Diagram

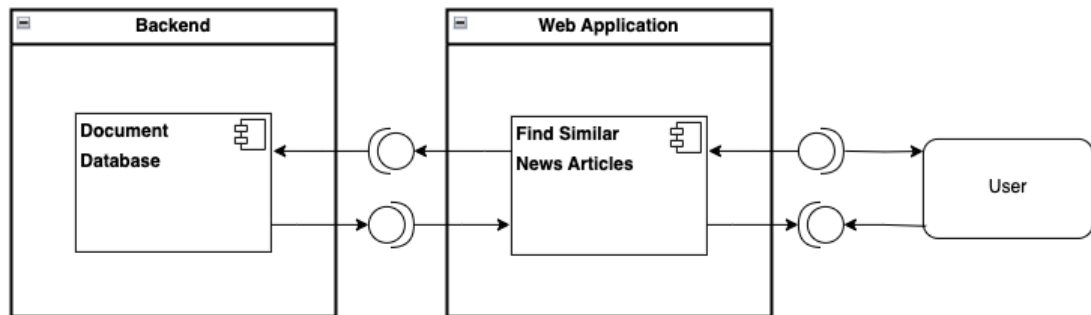


Figure 12: Component Diagram

Figure NUMBER is the component diagram illustrates the interaction between the user, web application, and the backend in the Khmer semantic similarity web application. This diagram shows the process of how user uses this application in a component level, consisting of module for finding the similar news articles (web application) which interacts with the document database (backend) and how the data

transaction between the three components. The user interacts with the system by inputting text data, which will be forwarded to the backend system for processing. After the model calculates the word embedding and compares the similarity between the input text and the text in the database, the backend returns the result to the frontend, which will be displayed to the user on the web application interface.

5.2.4 Activity Diagram

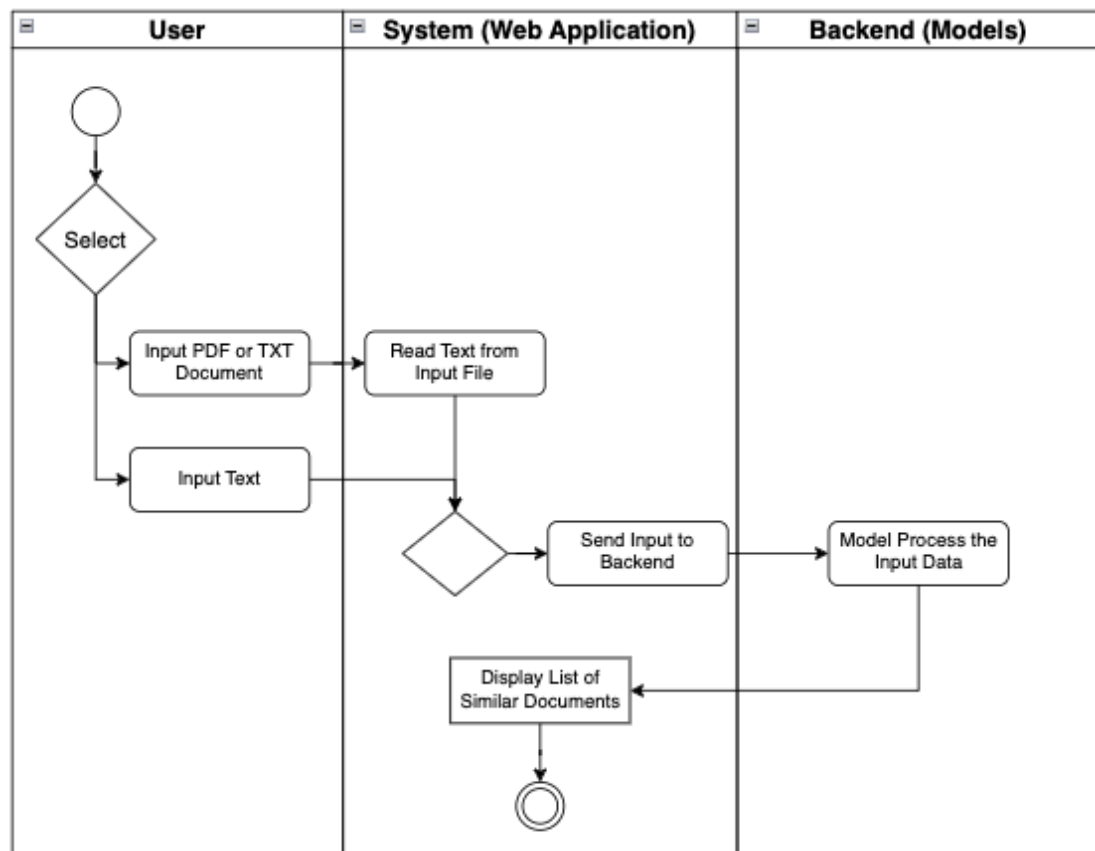


Figure 13: Activity Diagram

This figure NUMBER, is the activity diagram showing the workflow of the Khmer semantic similarity system. The user begins by choosing the word embedding models (TF-IDF, Fasttext) and a similarity measurement method (cosine similarity or Euclidean distance). Then they can choose whether to input a TXT document or input plain text straight to the web application. The system reads the input file or text and sends it to the backend for the model to calculate the word embedding vectors,

compare the similarity between the existing documents in the database and return a list of similar documents to the input text. The web application, which then displays the result back to the user. This work flow ensures efficient text analyzing and usage.

5.2.5 Roadmap User Stories

Epic	Module	User Stories	Acceptance Criteria
Main Page	Input document	As a user, I want to choose a TXT file to input	- user is provided with a file upload button to select a TXT file from their device
	Input text	As a user, I want to input text in the provided text box	- user provided with a text box to input text directly - the text box supports the basic text editing (copy, paste, edit)
	Obtain result	As a user, I want to see a list of documents that are similar to my input text	- the user is shown a loading indicator while system

Table 18: Roadmap User Stories

5.3 Software Project Management (SPMP)

5.3.1 Startup Plan

Project Communications

For communication during the timeline of this research, the researcher and the advisor utilize Telegram and Google Meet, in order to communicate and update

remotely. Moreover, Google Drive has been used for sharing files, documents, and source code.

Technical Process

Regarding the development of the system, this project follows Rapid Application Development (RAD) methodology. By utilizing the rad, both the development of the demo application and the development and the experimentation of the model. This approach allows for early changes in the development stage, accommodating user feedback

Tools

The project uses multiple tools that are categorized into three main sections, which are model development, web application (frontend) development, and backend development. Those tools are:

- Model development:
 - Programming language: Python
 - Machine learning framework:
 - Model training platform: Google Colab
 - Code editor: Visual Studio Code and Google Colab
 - Version control: GitHub
- Web application development:
 - Programming language: HTML, CSS, Javascript, Python
 - Web application framework: Jingja
 - Code editor: Visual Studio Code
 - Version control: GitHub

- Backend development:
 - Programming language: Python
 - Backend framework: Fastapi
 - Code editor: Visual Studio Code
 - Version control: GitHub

5.3.2 Work Plan

Activities and Tasks

All task and activities have done by the researcher.

Task	Task Description	Start Date	End Date
Redefine the Problem Statement		07/03/2024	10/03/2024
Make Changes to the Scope		11/03/2024	14/03/2024
Research on Related Literature	Read through existed literatures being done on semantic similarity	15/03/2024	21/03/2024
Research on literature on Khmer text	Research and read about semantic similarity research on Khmer language and on Khmer NLP	22/03/2024	28/03/2024
Research on Related Technologies	Go through and study about how related technologies on semantic similarity works	29/03/2024	04/04/2024
Research on Word Embeddings	Learn about how different types of word embedding works	05/04/2024	11/04/2024

Research on Data Requirement	Read about how different methods required different data format	12/04/2024	18/04/2024
Data Cleaning	Remove empty files, and make sure the dataset is clean for data preprocessing, define a naming format, change name	19/04/2024	22/04/2024
Data Preprocessing	Write functions to clean the content of the data, remove special character, not needed characters, and remove stop words	23/04/2024	26/04/2024
Set up Feature Extraction Process	Learn about and design the flow for each experimenting model	26/04/2024	26/04/2024
Setup Development Environment	Set up coding project for model development, and demo app development, set up GitHub repository for vision control	26/04/2024	26/04/2024
Build Baseline Model	Develop baseline model with TF-IDF word embedding and calculate the similarity using Cosine similarity and Euclidean distance	26/04/2024	10/05/2024
Set Decision Threshold for Baseline Model	Set a decision threshold on the calculation to ensure an optimal accuracy	26/04/2024	10/05/2024
Build FastText Model	Develop FastText model with FastText word embedding and calculate the similarity using Cosine similarity and Euclidean distance	10/05/2024	20/05/2024

Set up Experiment for FastText model	Set the model types and hyperparameters in configuration files for experimenting	20/05/2024	26/05/2024
Set Decision Threshold for FastText Model	Set a decision threshold on the calculation to ensure an optimal accuracy	20/05/2024	26/05/2024
Log Experiment	Log down results from each experiment	26/04/2024	05/06/2024
Build Graph for Visualization	Using the logs from the experiments to create graph and table	26/04/2024	05/06/2024
Design User Interface	Design a simple user interface for demonstrating the models	15/05/2024	20/05/2024
Build User Interface	Build the web application which is the user interface	15/05/2024	20/05/2024
Build Backend Server	Build a backend server to interact with the frontend	15/05/2024	03/06/2024

Table 19: Activities and Tasks

Release Plan

The research will be demonstrated on a prototype, therefore there will be release date as it will not be hosted and released. The source code to the models and the demo web application can be found on researcher's GitHub profile.

Iteration Plan

Iteration 1:

Goal:

- Set up development environment

Duration: 1 week

Objective:

- Setting up experimental pipeline
- Setting up model development environment and pipeline
- Setting up demo application environment

Iteration 2:

Goal:

- Model development and experiment

Duration: 4 week

Objective:

- Developing TF-IDF baseline model
- Developing FastText model
- Setting up demo application environment

Iteration 3:

Goal:

- Continue model development and experiment

Duration: 2 weeks

Objective:

- Experiments on TF-IDF
- Setting up demo application environment

Iteration 4:

Goal:

- Prototype development

Duration: 2 week

Objective:

- Designing the user interface
- Developing web application interface for user interaction
- Developing backend server

Budget

This project is research based project, therefore, the budget spent was on training and fine-tuning the large language model and hosting the web application which is for demonstrating the application.

Particulars	Item Details	Quantity	Units	Total Cost	Remarks
Computation Resource	Training Model (Google Colab)	1	\$10.42	\$10.42	For training

Table 20: Budget

5.3.3 Control Plan

Monitoring and Control

Daily	- Team meeting. Project team members report the progress and potential problems they have faced
Weekly	- Update
27/02/2024	- Revise the research proposal and present the research proposal to the panelists
29/02/2024	- Review and make changes to the proposal and the methods of the research
07/03/2024	- Start of development and experiment
15/05/2024	- Integration and testing
18/05/2024	- Final review

Project Measurements

The project measurements in this system are classified into planning, design, development, testing, deployment, and maintenance.

Phase	Measurement	Source
Planning	- Time taken to complete - Completeness of requirements, scopes, and goal	Researcher
Design	- Time taken to complete - Number of experimental methods and models	Researcher
Development	- Time taken to complete - Quality of code - Number of experimental methods and models	Researcher
Testing	- Time taken to complete - Number of bugs fixed - Number of experimenting configurations	Researcher
Deployment	- Time taken to complete - User rating	Researcher
Maintenance	Time taken to resolve	Researcher

Table 21: Project Measurements

5.3.4 Supporting Process Plan

Risk Management Plan

There are some potential risks in this project that are listed down in the table below:

ID	Risk	Contingency
R1	Delay in timeline	Start early and prepare and clean the dataset
R2	Model inaccuracy	Regularly update and evaluate the model, use diverse datasets for training, and implement feedback loop
R3	Integration issues with Supabase	Conduct thorough research and testing
R4	Development bugs	Have a clear version control history and experiments setup

Table 22: Risk Management Plan

Configuration Management Plan

1. Document: all documents and resources relating to this research are stored and shared in Google Drive. Change history can be reviewed.
2. Source code: the source code for the Khmer semantic similarity and the prototype for demonstration purpose are stored in a repository on GitHub.
3. Version Control: changes made to the source code on this research are monitored and can be reviewed in GitHub repository under history of commit.

Product Acceptance Plan

The product acceptance plan outlines the criteria and processes for evaluating the final product, and in this case, the demo application. The objective of this project is to research on different word embedding for the Khmer text and calculate the similarity using the vectors from the embedding models. The acceptance criteria are:

- Functional requirements: the system fulfills all the defined functional requirements which specified the functionalities of the system.
- Non-functional requirements: the system should fulfill the defined non-functional requirements, which includes the performance, security, and usability.
- Documentation: the system should comply with the written documents and the diagram describes the flow of the system.

Chapter 6: Summary of Findings, Conclusions, and Recommendations

6.1 Findings

In this study of word embedding models for Khmer semantic similarity, the data for training, testing, and evaluating are scraped from public websites on the internet such as BTV [21], Khmerload, Sabaynews, B8-sport, and Popular. Those raw news article data underwent processes such dataset creation, cleaning, and preprocessing, which eventually were used to test various of experiments. The experiments include dimensionality reduction on TF-IDF, evaluation and comparison between TF-IDF and FastText, as well as comparison between similarity measurements (Cosine Similarity and Euclidean Distance).

In the training of TF-IDF, a number of experiments on dimensionality reduction were tested on, and below is the result from those experiments. It is categorized into types of measurements.

TF-IDF					
Type: Dimensional Reduction					
Similarity Measurement: Cosine Similarity					
Decision Threshold = 0.5					
Experiment	Accuracy	Precision	Recall	F1-Score	Dimension
0: min_df=0	76.1%	100%	52.3%	68.7%	48796

1: min_df=0.01	83.7%	100%	67.5%	80.5%	1636
2: min_df=0.02	84.9%	100%	69.8%	82.2%	923
3: min_df=0.1	87.7%	97.3%	77.6%	86.3%	145
4: min_df=0.2	82.3%	82%	82.7%	82.4%	46
Similarity Measurement: Euclidean Distance					
Decision Threshold = 1.3					
Experiment	Accuracy	Precision	Recall	F1-Score	Dimension
0: min_df=0	98.7%	99.8%	97.6%	98.7%	48796
1: min_df=0.01	98%	97.5%	98.6%	98%	1636
2: min_df=0.02	95.7%	93.6%	98.1%	95.8%	923
3: min_df=0.1	73.3%	65.9%	96.4%	78.3%	145
4: min_df=0.2	60.3%	56%	95.5%	70.6%	46

Table 13: Dimensionality Reduction Experiments

Moreover, both TF-IDF and FastText models are evaluated on the pair dataset that was created that consists of similar and dissimilar pairs. The result from FastText has a better performance of 94.5% accuracy rate and 94.3% f1-score, and TF-IDF has

the accuracy rate of only 87.7% and f1-score of 86.3%. As compare to the highest performance from the TF-IDF, FastText model uses vector of 100 dimension to represent the terms, where TF-IDF uses 145 dimensions.

Word embedding	Configuration	Similarity measurement	Accuracy Score	Precision	Recall	F1 score
TF-IDF	-Decision threshold: 0.5, -min_df=0.1	Cosine Similarity	87.7%	97.3%	77.6%	86.3%
	-Decision threshold: 1.3 -min_df=0.01	Euclidean Distance	98%	97.5%	98.6%	98%
FastText	Decision threshold: 0.1	Cosine Similarity	94.5%	97.8%	91.1%	94.3%
	Decision threshold: 0.5	Euclidean Distance	82.6%	99.8%	65.3%	78.9%

Table 14: Comparison between TF-IDF and FastText

6.2 Conclusion

According to the findings of this study, the results led to the following conclusions:

1. Dimensionality reduction can affect the performance of the model. The data from the experimentation has shown that the performance of TF-IDF reaches its peak when the vector representation has 145 dimension, whereas the accuracy before this reduction is only 76.1%. Based on this data, dimensionality reduction really do has an impact on the performance.

2. FastText model performs better than the static model TF-IDF. Based on the testing and evaluation, FastText is found to have the accuracy rate of 94.5% with the dimension of 100, whereas for TF-IDF, reached its performance at 87.7% with the dimension of 145. Therefore, FastText model has a higher accuracy.
3. Word representation can be utilized for text comparison by giving vector values to each term in the documents of the dataset. Next, similarity measurement such as Cosine Similarity or Euclidean Distance is utilized to compute the distance, which can be used to classify for its similarity. In this case, FastText model and Cosine Similarity will be used to demonstrate its performance through a web application.

6.3 Recommendation

Given the findings resulted from various experiments, there are still a various space for improvement and adjustment, both to the models and the prototype. The recommendations are as listed down below:

- As per the demo web application, the future development can include larger file upload that can be extracted and used for computing for the similar documents
- As for future research, the experimentation can be explored to BERT or GPT for word embedding, relating to the performance on Khmer language and the accuracy
- As for future research, the researchers can try using BERT classification head for classifying task and evaluate the performance
- Regarding the similarity classification, future researchers can look for other

factors that may affect the similarity measurement, and factors that can improve the performance of the embedding model

6.4 Challenges

Translation on the dataset

In the process of creating the dataset for this similarity task, pairs of documents are needed for evaluating the model. A pair of a document is a paraphrase of the original document, written in different ways, yet have similar meaning or topic. In order to obtain such files, two translation steps are performed, which is the back translation. This process takes a significant amount of time as each document is Khmer news articles ranging from short to moderate, and to longer length.

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Appendices

Appendix A: Researcher Curriculum Vitae



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Education	Paragon International University Bachelor in Computer Science	expected: 27th July 2024
	TUX Global Institute App/Web Development Course	2021 - 2022
	Australia Centre for Education General English Program	2015 - 2019
	Goethe-Zentrum Phnom Penh General German Language	2023 - Present

Work	ML Engineer Intern EKYC Solutions	Aug 2023 - Apr 2024
Experience	Developed application for detecting authentic ID card document with image processing	

Data Analyst Intern TUX Global Institute Worked with PowerBI for dashboard and report	Dec 2022 - Apr 2023
Web Admin Intern Tech for Kids Academy Managed data and content for the academy's website, improved the usability and performance	May - Oct 2022
Facilitator Snoopedu Responsible for four classes grade 69, taught scratch programming, MakeCode, and microbit	Jul - Sep 2022
Facilitator She Can Code, Paragon University Instructed 10 students on coding assignments, give score, design weekly posters and videos	Sep - Oct 2021
Facilitator WESTEAM Facilitated students from Battambang Province online, on microbit and Scratch programming	Nov 2021 - Apr 2022

Language	Khmer	Native
	English	Fluent
	German	Beginner

Personal Information	Age	22
	Gender	Female
	Nationality	Khmer

Appendix B: Product Backlog

List of Features			1 Story point = 4 hours		
Feature ID	Epics	Description	Story Point	Status	Remark
1	Redefine the Problem Statement	Modified according to the recommendations	1	Done	
2	Make Changes to the Scope	Modified according to the recommendations	1	Done	
3	Research on Related Literature	Read through existed literatures being done on semantic similarity	15	Done	
4	Research on literature on Khmer text	Research and read about semantic similarity research on Khmer language and on Khmer NLP	15	Done	
5	Research on Related Technologies	Go through and study about how related technologies on semantic similarity works	8	Done	
6	Research on Word Embeddings	Learn about how different types of word embedding works	10	Done	
7	Research on Data Requirement	Read about how different methods required different data format	3	Done	
8	Data Cleaning	Remove empty files, and make sure the dataset is clean for data preprocessing, define a naming format, change name	12	Done	
9	Data Preprocessing	Write functions to clean the content of the data, remove special character, not needed	3	Done	

		characters, and remove stop words			
10	Set up Feature Extraction Process	Learn about and design the flow for each experimenting model	3	Done	
11	Setup Development Environment	Set up coding project for model development, and demo app development, set up GitHub repository for vision control	2	Done	
12	Build Baseline Model	Develop baseline model with TF-IDF word embedding and calculate the similarity using Cosine similarity and Euclidean distance	6	Done	
13	Set Decision Threshold for Baseline Model	Set a decision threshold on the calculation to ensure an optimal accuracy	3	Done	
14	Build FastText Model	Develop FastText model with FastText word embedding and calculate the similarity using Cosine similarity and Euclidean distance	8	Done	
15	Set up Experiment for FastText model	Set the model types and hyperparameters in configuration files for experimenting	3	Done	
16	Set Decision Threshold for Fasttext Model	Set a decision threshold on the calculation to ensure an optimal accuracy	2	Done	
17	Log Experiment	Log down results from each experiment	2	Done	
18	Build Graph for Visualization	Using the logs from the experiments to create graph and table	2	Done	
19	Design User	Design a simple user interface	3	Done	

	Interface	for demonstrating the models			
20	Build User Interface	Build the web application which is the user interface	5	Done	
21	Build Backend Server	Build a backend server to interact with the frontend	6	Done	

Appendix C: Experimental Setup and Result

TF-IDF					
Type: Dimensional Reduction					
Similarity Measurement: Cosine Similarity					
Decision Threshold = 0.5					
Experiment	Accuracy	Precision	Recall	F1-Score	Dimension
0: min_df=0	76.1%	100%	52.3%	68.7%	48796
1: min_df=0.01	83.7%	100%	67.5%	80.5%	1636
2: min_df=0.02	84.9%	100%	69.8%	82.2%	923
3: min_df=0.1	87.7%	97.3%	77.6%	86.3%	145
4: min_df=0.2	82.3%	82%	82.7%	82.4%	46
Similarity Measurement: Euclidean Distance					
Decision Threshold = 1.3					
Experiment	Accuracy	Precision	Recall	F1-Score	Dimension
0: min_df=0	98.7%	99.8%	97.6%	98.7%	48796
1: min_df=0.01	98%	97.5%	98.6%	98%	1636

2: min_df=0.02	95.7%	93.6%	98.1%	95.8%	923
3: min_df=0.1	73.3%	65.9%	96.4%	78.3%	145
4: min_df=0.2	60.3%	56%	95.5%	70.6%	46

Table 13: Dimensionality Reduction Experiments

Appendix D: Endorsement Letters



ENDORSEMENT LETTER

In partial fulfilment for the requirements for the degree Bachelor of Science in Computer Science. This thesis entitled **"ENHANCING KHMER TEXTUAL SIMILARITY ASSESSMENT: ALGORITHMIC SYNERGY FOR IMPROVED MODEL ACCURACY"** has been examined and recommended for Pre Oral Defense.

Chamroeun Khim
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A blue ink signature of Chamroeun Khim, written in a cursive style, enclosed within a rectangular box.

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ENDORSEMENT LETTER

In partial fulfilment for the requirements for the degree Bachelor of Science in Computer Science. This thesis entitled **"ENHANCING KHMER TEXTUAL SIMILARITY ASSESSMENT: ALGORITHMIC SYNERGY FOR IMPROVED MODEL ACCURACY"** has been examined and recommended for Pre Oral Defense.

Socretquuliqaa Lee
ENGLISH CRITIC

A handwritten signature in blue ink, appearing to be "Socretquuliqaa Lee", written over a large, faint, stylized graphic of three vertical lines of increasing height.

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Appendix D: Certificates of Digital Signature